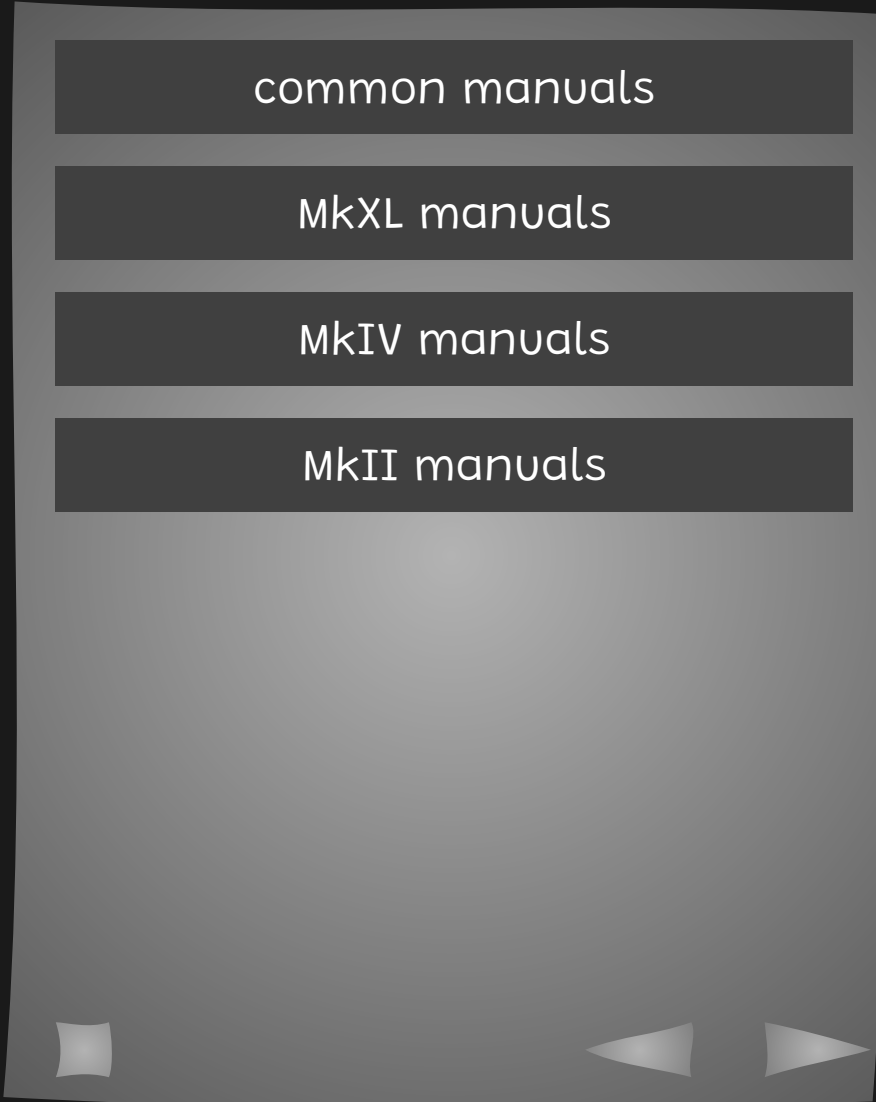




common manuals

MkXL manuals

MkIV manuals

MkII manuals

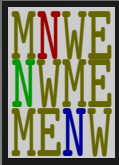
## The manuals

Welcome to the suite of ConT<sub>E</sub>Xt manuals. These manuals not only cover the macro package itself, but also the tools that come with it. In this suite you will also find manuals on how to use ConT<sub>E</sub>Xt for processing xml. Fonts and MetaPost graphic are discussed in dedicated manuals.

On the following pages, the main manuals are shown large, while their screen companions are shown in the bottom right corner of a page. Clicking on a picture brings you to the document. Some manuals come in more than one language, in which case small pictures of the title pages are shown.

The next pages show overviews of manuals that are specific for MkII, MkIV and , as well as obsolete manuals. The archived manuals cover MkII but can still be of interest, if only because they give an idea about how all evolved.

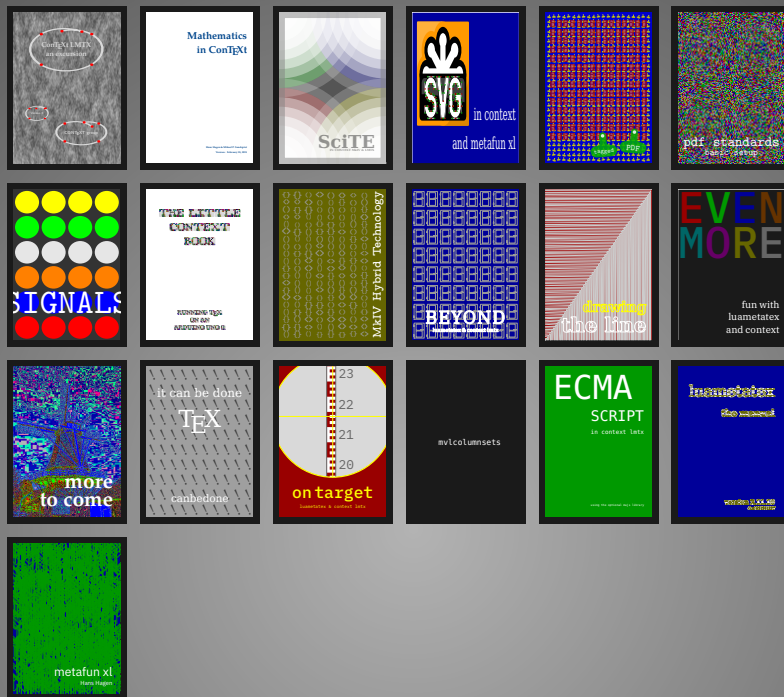
This is a snapshot. Check the distribution for the latest versions and extra manuals.



# Common manuals

These are generic manuals. The more version specific manuals are collected on the following pages, including some historic ones. In addition to the manuals you can have a look at the Wiki and some manuals (in various languages) by users that can be found on the Internet.





# MkXL manuals

All the good things come together in the LMTX where we pair LuaMetaTeX with MkXL. This combination is more powerful and performs better. Most of the MkIV manuals apply but there are some that are more exclusive to the latest greatest. Manuals tagged at MkIV might not compile with LuaMetaTeX due to new features but still be usefull for that version.



# MkIV manuals

Here you will find the manuals that describe MkIV functionality and/or features not present in MkII. But again we froze a version when we moved on, this time to LuaMetaTeX. The MkIV version is the test-case for LuaTeX so it will adapt when needed.

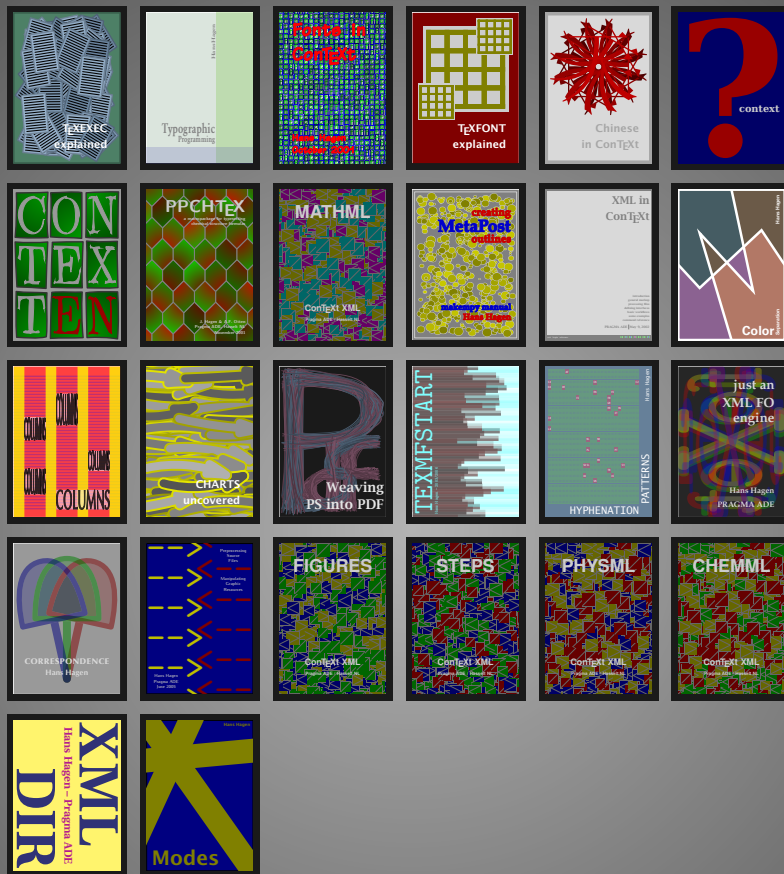




# MkIV manuals

Here you will find the manuals that describe MkIV functionality and/or features not present in MkII. But again we froze a version when we moved on, this time to LuaMetaTeX. The MkIV version is the test-case for LuaTeX so it will adapt when needed.

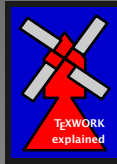
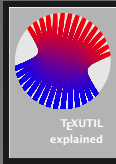
(continued)



# MkII manuals

Although MkII and MkIV are rather compatible, there are some differences. Also, as MkII is frozen there are more features MkIV. Also because we moved to Unicode and OpenType there are differences in those areas.

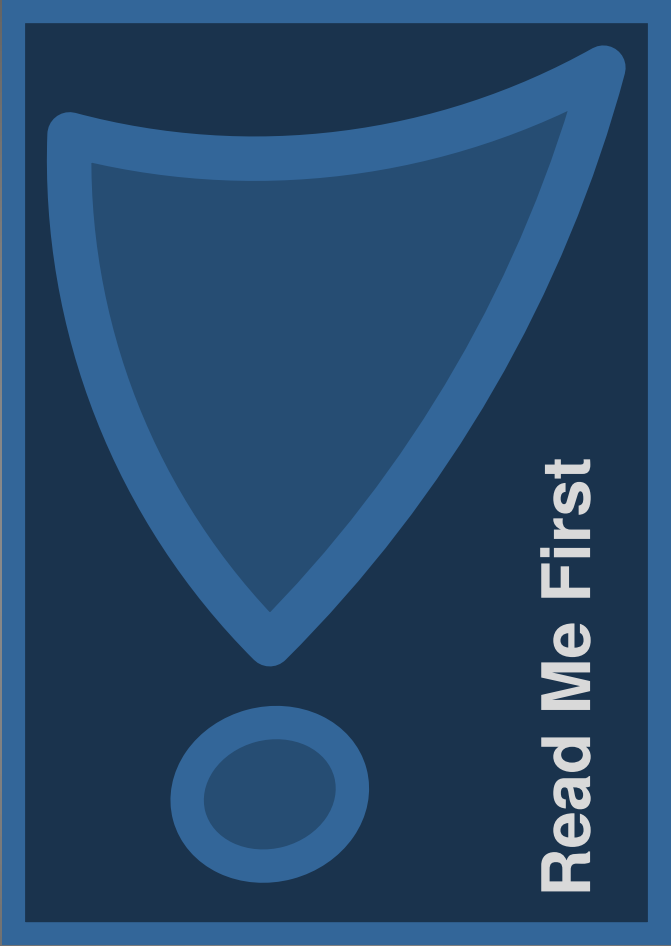
When you want to process these manuals you need a MkII installation which includes fonts no longer present in the later versions. We will not update these manuals! They describe history.



# Obsolete manuals

We keep some of the old manuals around for historic reasons. Some of what is described might still float around in the distribution but is likely replaced by more modern and hip variants.





Read Me First

## Read Me

It's in the name: you should read this file. Not so much because the content should bother you, but more because it gives you an idea about what we have in mind with making ConT<sub>E</sub>Xt available for everyone.

ConT<sub>E</sub>Xt is completely free software, which does not mean that there are no restrictions on redistributing and changing the files. When you want to redistribute (changed) source code, please read this licence first.



# ConTeXt

## commands

# EN

## Quick References

This quick reference manual does not replace the other manuals, but advanced users can benefit from its compactness. The manual can be generated on the user's system, since the style and database that is needed is part of the distribution.



setup.cz





## MWE

When you run into troubles the best place to get help is the mailing list. There the magic word is .

# What is ConT<sub>E</sub>Xt

This is a one page summary of what ConT<sub>E</sub>Xt is about.

There are several ways to deploy ConT<sub>E</sub>Xt. Most common is to let it render a document, in which case you install it on a system and use an editor to input your document and from a console or by clicking some key trigger a run. The input can be structured using T<sub>E</sub>X macros but it can also be xml, or some other format that gets converted to ConT<sub>E</sub>Xt commands before processing.

Instead you can also use ConT<sub>E</sub>Xt as a more hidden application, for instance in a web service or rendering component in a larger application. In that case the end user is not really aware that T<sub>E</sub>X is being used.

No matter how you use ConT<sub>E</sub>Xt, you need to install it first. You can for instance use T<sub>E</sub>Xlive or another distribution to pick up ConT<sub>E</sub>Xt, but you can also install it using the archive (snapshot), in which case you also need to pick up the engine (for instance LuaT<sub>E</sub>X) and a basic set of fonts. In order to make installation easy we provide a so called standalone distribution that has all you need.

The standalone ConT<sub>E</sub>Xt distribution has the following characteristics:

- The installation is self contained. Apart from resources like fonts, the T<sub>E</sub>X macros, Lua code and MetaPost helpers are provided in one package.
- There is only one binary involved: LuaMetaT<sub>E</sub>X. The source code of this program is integral part of the ConT<sub>E</sub>Xt distribution (per end 2019). A user should be able to compile the program if needed. There is no dependency on additional libraries other than those that make up the operating system.
- The core system is able to typeset documents in an efficient way. The memory footprint is decent and performance acceptable, also on low power devices and virtual machines. We try not to provide bloatware.
- The official user interface is stable and the implementation of core components will not change fundamentally. When something can be improved it will be. One can use a snapshot for long term stability.
- Support is provided by means of a mailing list, a wiki, documentation, meetings, etc. If needed you can consult (or hire) support. There are enough experienced users out there to get you going.

The first version of ConT<sub>E</sub>Xt, now tagged MkII, has been around since 1995 and (still) runs on top of pdfT<sub>E</sub>X. The development of its successor MkIV started in 2005 as part of the LuaT<sub>E</sub>X development and still carries on. The most recent incarnation is lmtx, which is MkIV but tuned for LuaMetaT<sub>E</sub>X, the lean and mean successor of LuaT<sub>E</sub>X.

## ConT<sub>E</sub>Xt lmtx — factsheet

The LuaMetaTeX engine is a follow up on LuaTeX. It integrates the TeX text rendering engine, the MetaPost graphic engine and the Lua script interpreter. The development is part of the ConTeXt macro package development. This macro package tightly integrates the three subsystems. The LuaMetaTeX code is part of the ConTeXt distribution.

The LuaMetaTeX engine is lean and mean. There is for instance no backend code present. In ConTeXt this is handled in Lua. Graphic inclusion is also delegated to Lua. The TeX frontend is a slightly extended version of the LuaTeX one. System dependencies are minimized. Where possible we stay close to the original TeX concept because that is a well documented reference. The binary can also be used as stand alone Lua engine.

The MetaPost library also has access to Lua, which means that the language can be enhanced and functionality added on demand. There are several graphic libraries provided in ConTeXt. This graphical language is efficient in runtime and graphical output. In combination with Lua we have a high performance graphical subsystem that can handle a huge amount of data. Additional text (like labels) is typeset at high quality.

The Lua code that comes with ConTeXt contains a lot of helper code which means that one can set up selfcontained workflows without many extra dependencies. Documents can be encoded in TeX, Lua, xml or whatever suits. There is support for databases too.

The ConTeXt code base evolved over time. The basic functionality is quite stable. The move from MkII to MkIV to lmtx has been gradual. The efficiency in terms of code and performance has been improved stepwise. Development continues and beta releases occur on a regular basis. The ConTeXt user community is quite willing to experiment with betas that can be installed alongside stable snapshots.

The installation is relatively small (fonts make up much of it) and updating is easy. We operate in the TeX Directory Structure, which is a proven concept.

LuaMetaTeX — factsheet

# What is LuaMetaTeX

This is a one page summary of what LuaMetaTeX is about.

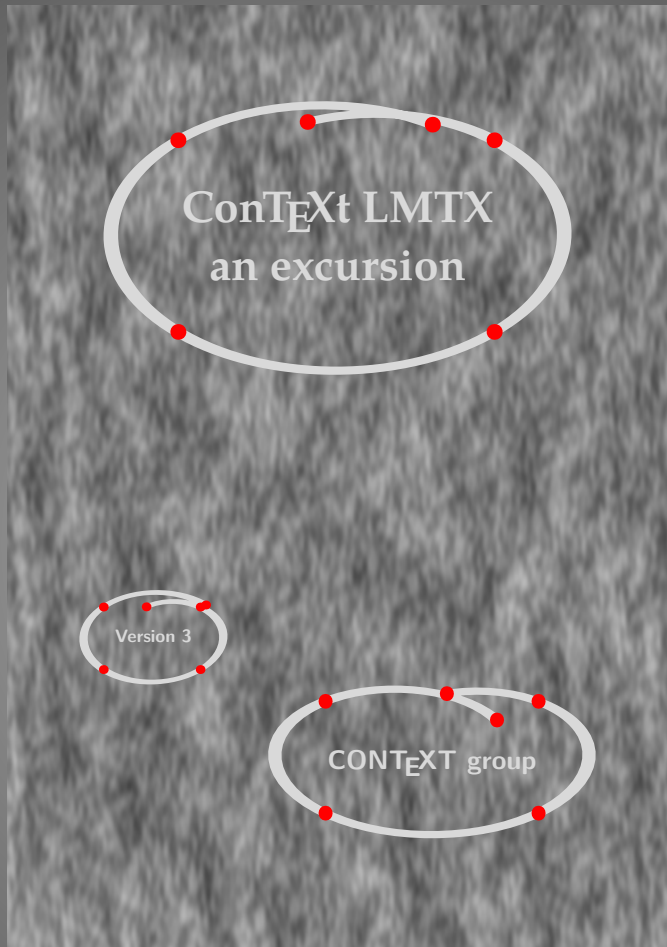
# Input methods

These few pages show various input methods in ConTeXt. There are more so reading might trigger you to look at various manuals, articles and examples.



# Excursion

This is the follow up on the getting started manual. It gives a overview of the most important ConT<sub>E</sub>Xt commands. The older manual covers MkII and MkIV.



# Mathematics in ConT<sub>E</sub>Xt

Hans Hagen & Mikael P. Sundqvist  
Version: February 28, 2025

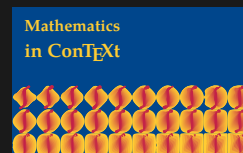
## Math In ConT<sub>E</sub>Xt

This manual describes how to typeset math in ConT<sub>E</sub>Xt LMTX. It discusses the various mechanisms but also shows how to code without tweaking in order to keep a clean source. It uses the advanced math rendering options available in LuaMetaT<sub>E</sub>X and has numerous examples.



examples-

matheanings





## SciTE in ConT<sub>E</sub>Xt

SciTE is an editor and these manuals describe how to configure it for use with ConT<sub>E</sub>Xt and MetaFun. Beware, the mscite manuals are the old ones, still valid for traditional lexing, while the readme version describes the latest greatest lexers.

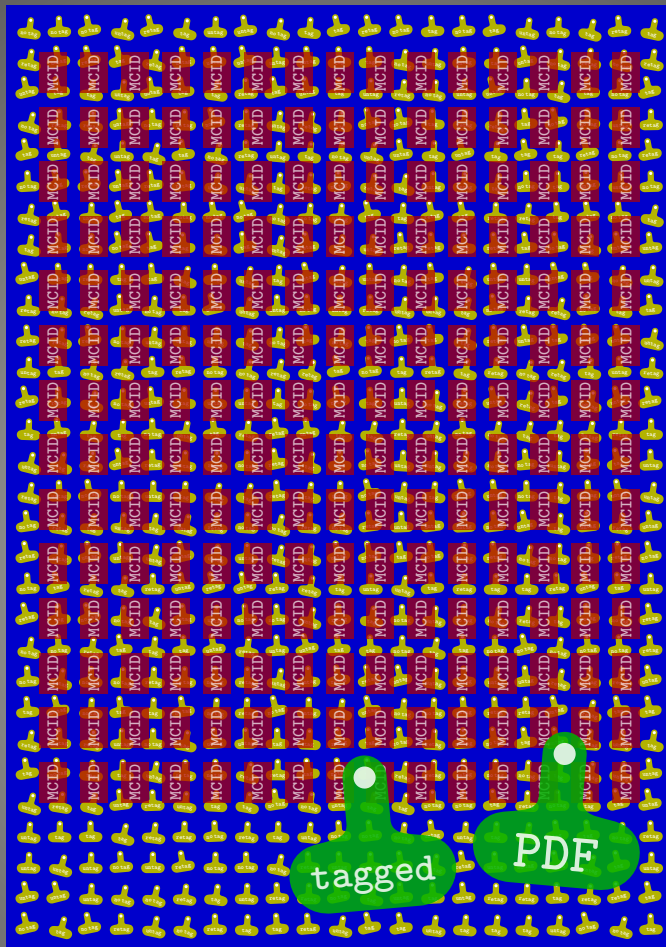
# SVG

You can let ConT<sub>E</sub>Xt convert svg images. It does so with the help of MetaPost.



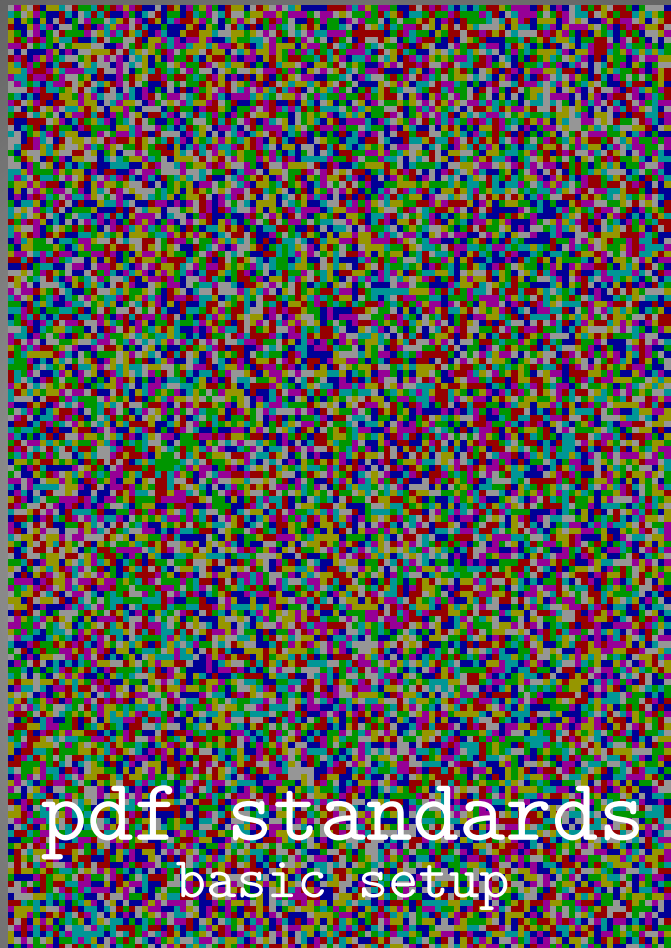
in context  
and metafun xl





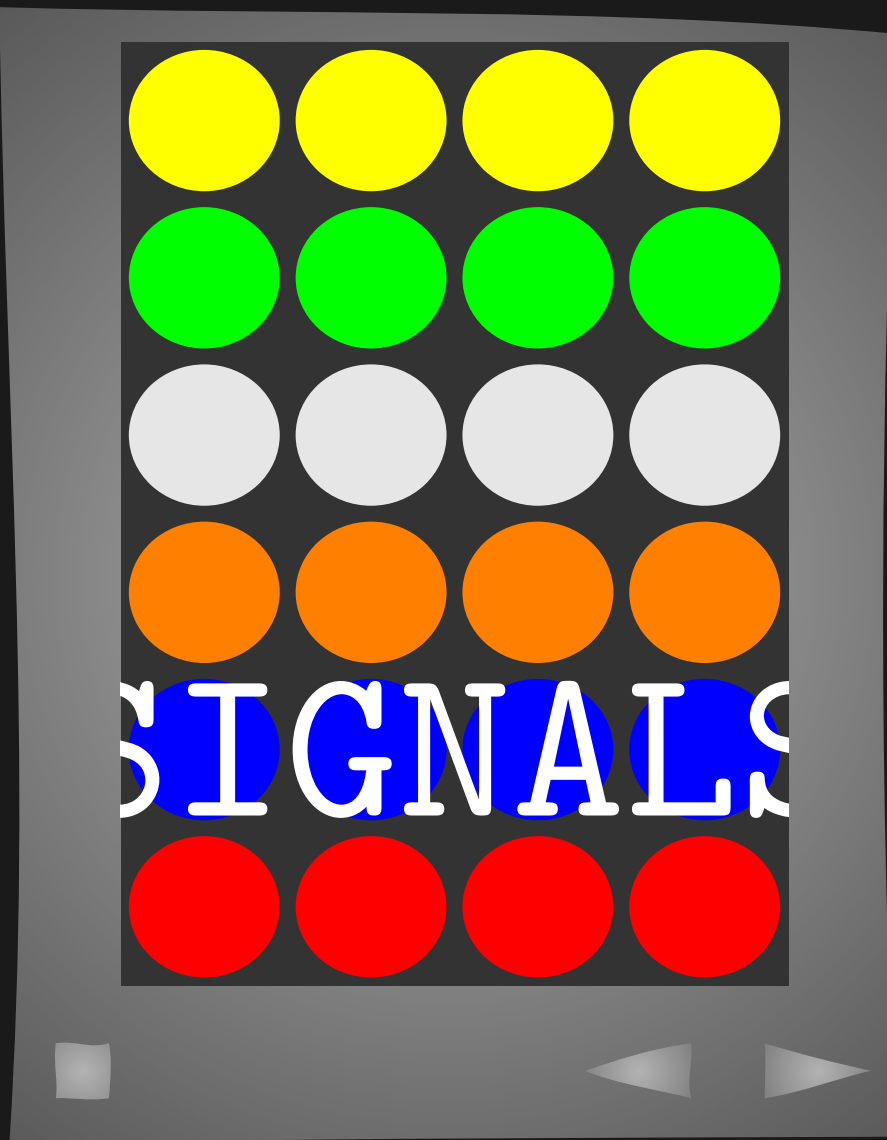
# Tagging

When you are forced to provide tagged pdf  
this manual is your friend.



# Standards

In case you're forced to use pdfstandards, this manual can be of help.



# Signals

This manual introduces you to the signal progress and error feature assisted by the 'ConT<sub>E</sub>Xt watch' presented at the 2025 meeting. We're talking of a gadget here.

# THE LITTLE CONTEXT BOOK

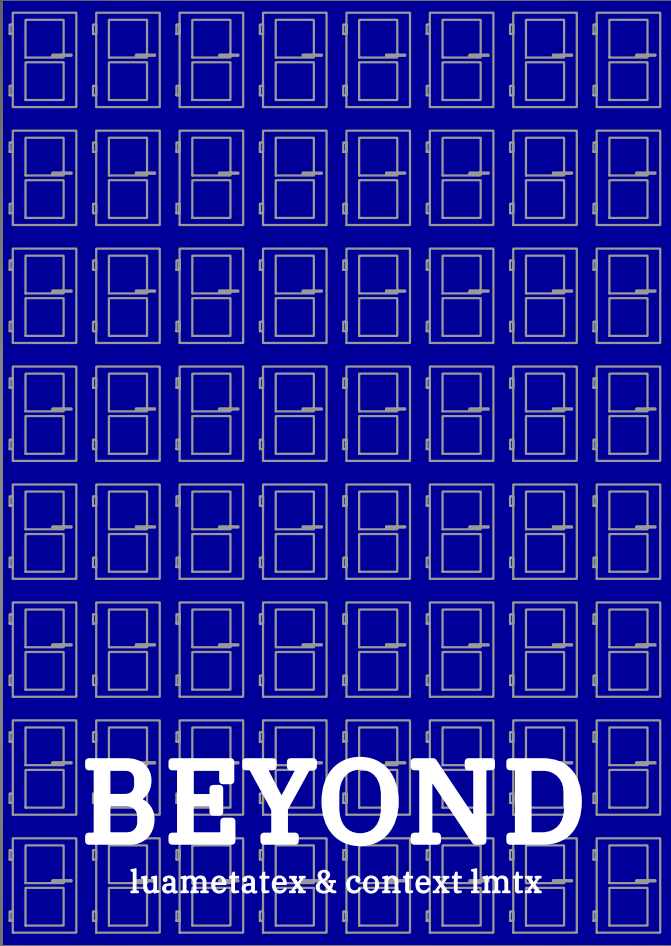
RUNNING T<sub>E</sub>X  
ON AN  
AROUND UNO 0

## Little

This little manual introduces the ‘little context book’ that we presented at BachoT<sub>E</sub>X 2026 as statement against the energy and environmental side effects of artificial intelligence (datacenters).

This document is one in the series that keeps track of various developments in Lua-TeX, LuaMetaTeX, ConTeXt and LuaMetaFun.

Here you can also find explanations not found in manuals. As with earlier development stories, some were written as articles for user group journals.



# BEYOND

luametatex & context lmtx

## MkXL development: beyond

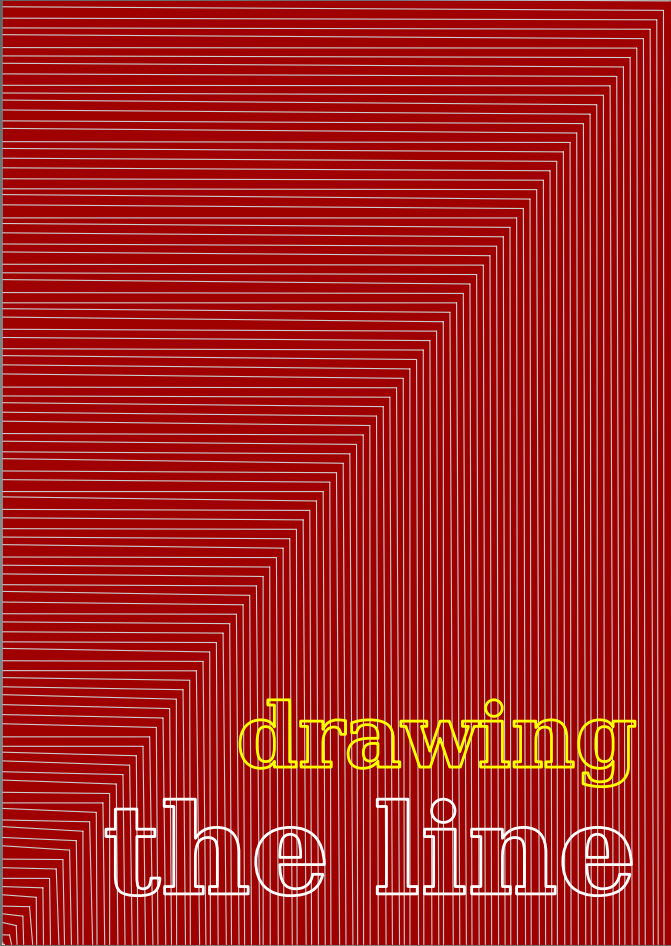
This document is one in the series that keeps track of various developments in Lua<sub>T</sub><sub>E</sub>X , LuaMeta<sub>T</sub><sub>E</sub>X, Con<sub>T</sub><sub>E</sub>Xt and LuaMetaFun.

Here you can also find explanations not found in manuals. As with earlier development stories, some were written as articles for user group journals.

# MkXL development: drawing lines

This document is one in the series that keeps track of various developments in Lua<sub>T</sub><sub>E</sub>X , LuaMeta<sub>T</sub><sub>E</sub>X, Con<sub>T</sub><sub>E</sub>Xt and LuaMetaFun.

Here you can also find explanations not found in manuals. As with earlier development stories, some were written as articles for user group journals.



drawing  
the line



EVEN  
MORE

fun with  
luametatex  
and context

## MkXL development: even more

This document is one in the series that keeps track of various developments in Lua<sub>T</sub><sub>E</sub>X , LuaMeta<sub>T</sub><sub>E</sub>X, Con<sub>T</sub><sub>E</sub>Xt and LuaMetaFun.

Here you can also find explanations not found in manuals. As with earlier development stories, some were written as articles for user group journals.



## MkXL development: more to come

This document is one in the series that keeps track of various developments in Lua<sub>T</sub><sub>E</sub>X , LuaMeta<sub>T</sub><sub>E</sub>X, Con<sub>T</sub><sub>E</sub>Xt and LuaMetaFun.

Here you can also find explanations not found in manuals. As with earlier development stories, some were written as articles for user group journals.





it can be done

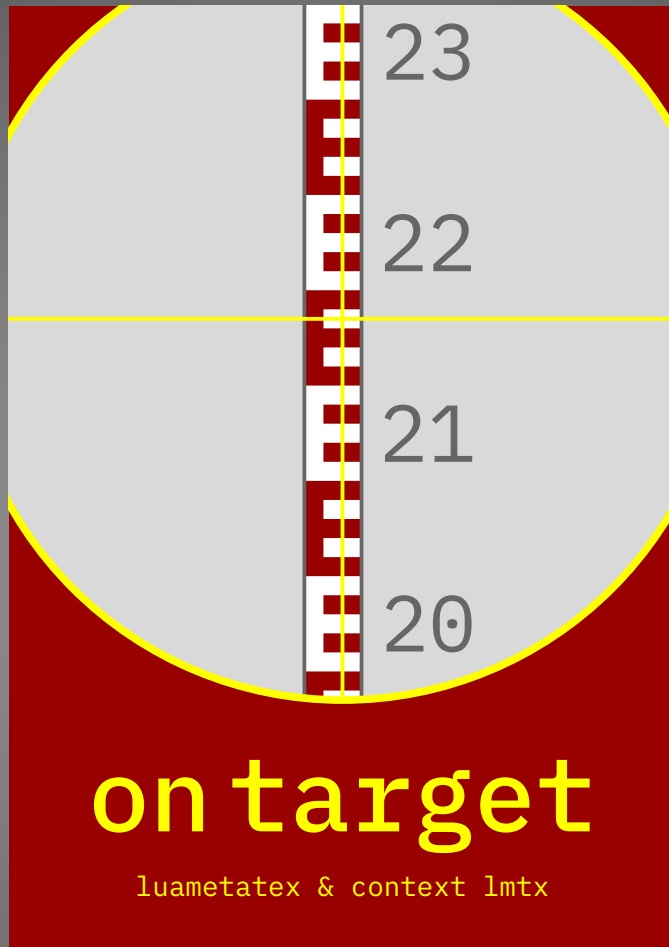
TEX

canbedone

## MkXL development: can be done

This document is one in the series that keeps track of various developments in Lua-TeX, LuaMetaTeX, ConTeXt and LuaMetaFun.

Here you can also find explanations not found in manuals. As with earlier development stories, some were written as articles for user group journals.



## MkXL development: on target

This document is one in the series that keeps track of various developments in Lua<sub>T</sub><sub>E</sub>X , LuaMeta<sub>T</sub><sub>E</sub>X, Con<sub>T</sub><sub>E</sub>Xt and LuaMetaFun.

Here you can also find explanations not found in manuals. As with earlier development stories, some were written as articles for user group journals.

# Columnsets

The new columnset implementation does whole-set optimization, is more robust and efficient and also more powerful. This manual explains how to define and populate pages and create parallel versions of documents. This is work in progress.

```
name: documents/general/manuals/mkx1/mv1columnsets.pdf
file: documents/general/manuals/mkx1/mv1columnsets.pdf
state: unknown
```

# ECMA SCRIPT

in context lmtx

using the optional mujs library

## ECMAScript in Con- TeXt LMTX

It is possible to embed ecmascript (of which JavaScript is an example) in ConT<sub>E</sub>Xt LMTX. The (external) library used is .

# luametatex

## the manual

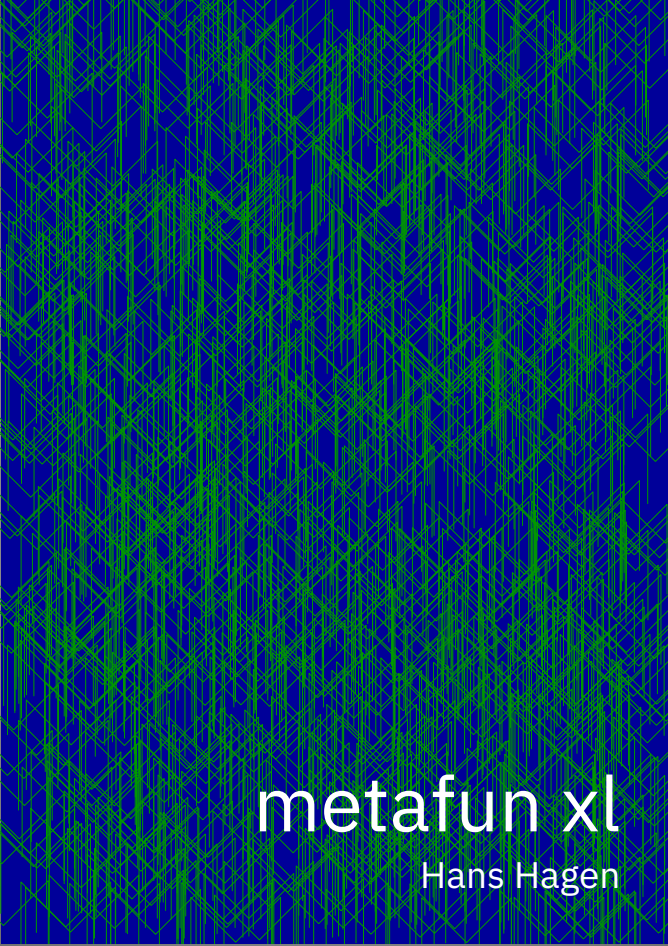
version 2.11.10

dev id 20260917

## LuaMetaTeX

The LMTX version of ConT<sub>E</sub>Xt uses the LuaMetaT<sub>E</sub>X engine. This engine is a follow up on LuaT<sub>E</sub>X and is an integral part of ConT<sub>E</sub>Xt development. This manual describes the current (official) features.

The related manual about primitives discusses some of the new primitives that relate to low level macro programming.



metafun xl

Hans Hagen

# LuaMetaFun

In LuaMetaTeX we use an enhanced version of MetaPost that is supported by extensions to MetaFun discussed in this manual.

There is a dedicated manual for 3D graphics that describes in great detail how to generate implicit, parametric, and related surface plots in high quality bitmap format bitmaps.





# ConTeXt Mark IV

## an excursion



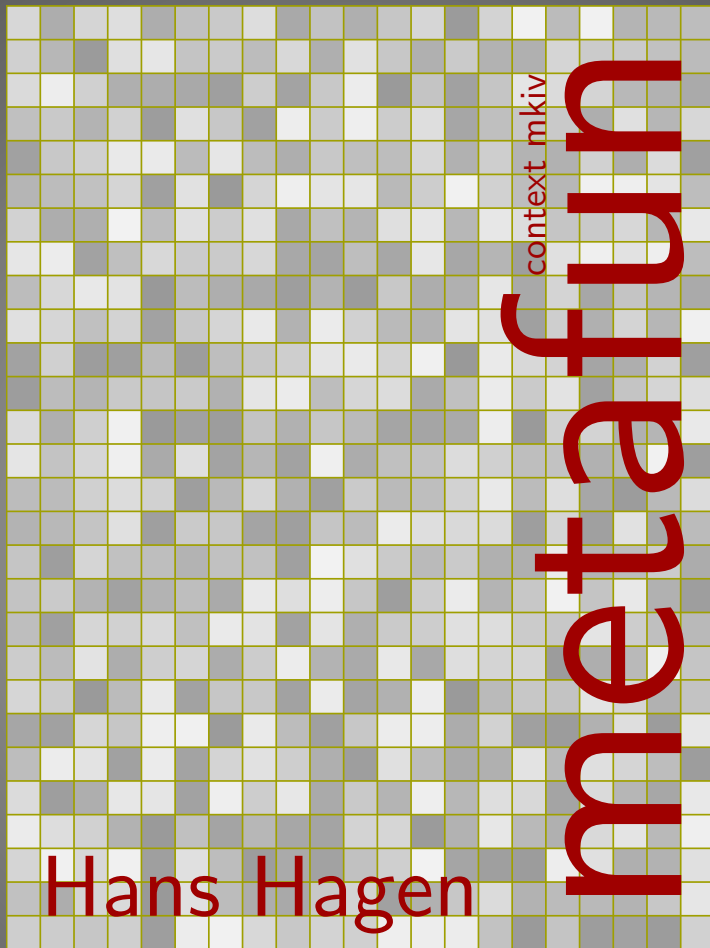
English



Ton Otten  
PRAGMA ADE

## Getting started

Although meant for beginners, these manuals shows already a lot of what ConTeXt can do for you. They also demonstrate that T<sub>E</sub>X documents can be colorful and can contain lots of graphics. See the excursion manual for an update!



# MetaFun

If you like graphics, you may like MetaFun, a collection of MetaPost macros. The manual covers most of MetaPost, as well as the interface between this graphical environment and ConT<sub>E</sub>Xt. There are numerous examples, that give you an impression about the power of this graphical system as well as the strength of the combination with T<sub>E</sub>X.



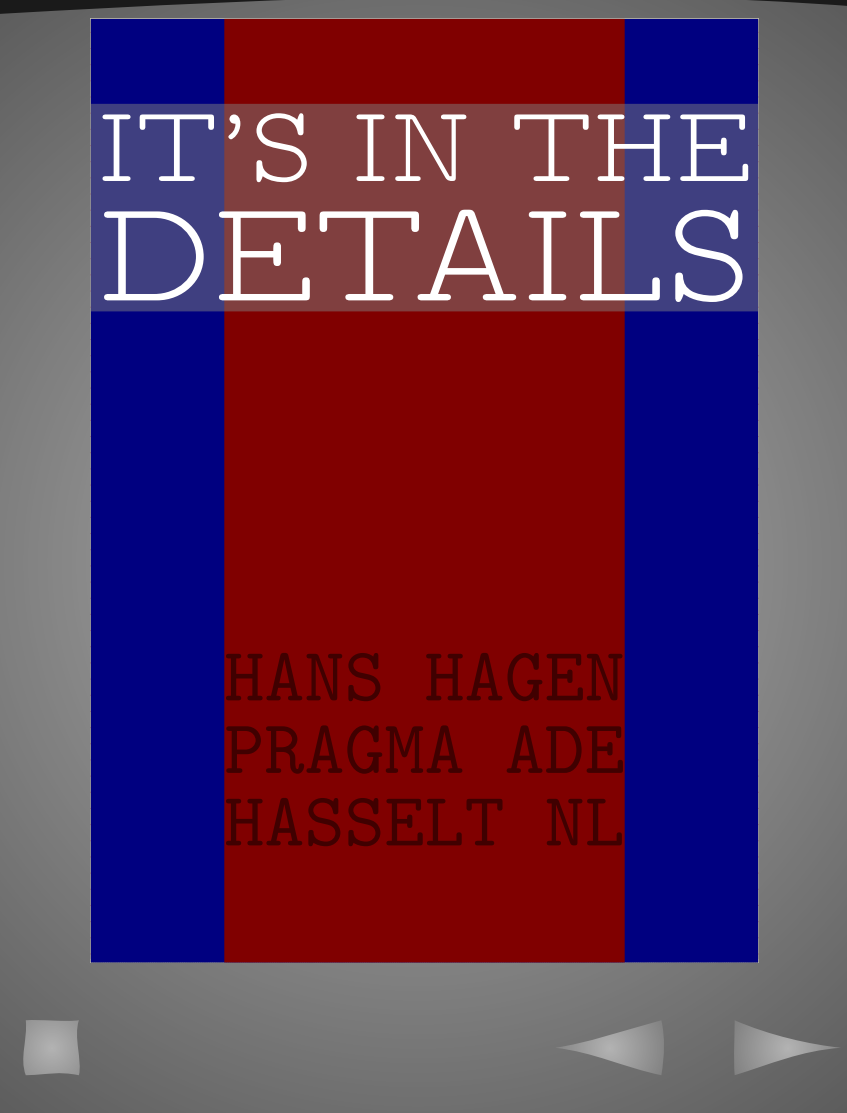




# Interaction

Producing interactive documents have always been an integral part of ConT<sub>E</sub>Xt. This manual describes how to configure hyperlinks, comments, attachments, forms and also how to add navigational elements to a document designed for display.





# IT'S IN THE DETAILS

HANS HAGEN  
PRAGMA ADE  
HASSELT NL

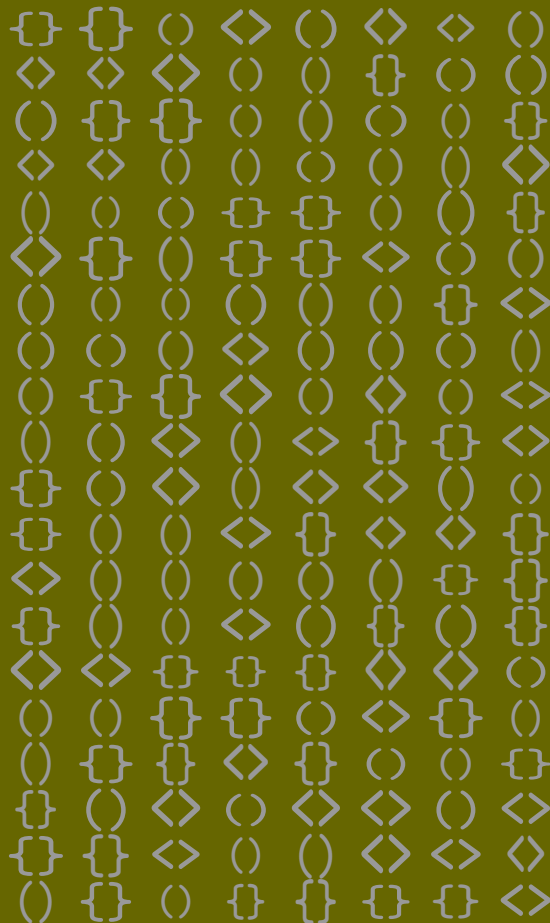
## It's in the details

This manual is meant for users who want to divert from the more or less traditional looking T<sub>E</sub>X documents. There is a strong focus on elements that determine the look and feel of a document, like graphics. (This manual is unfinished)



# ConT<sub>E</sub>Xt MkII – MkIV, the history of LuaT<sub>E</sub>X

This document keeps track of the development history of both ConT<sub>E</sub>Xt (mkiv) as well as LuaT<sub>E</sub>X. It is also one of our torture tests for both (rather related) systems.



# MkIV Hybrid Technology

## MkIV hybrid technology

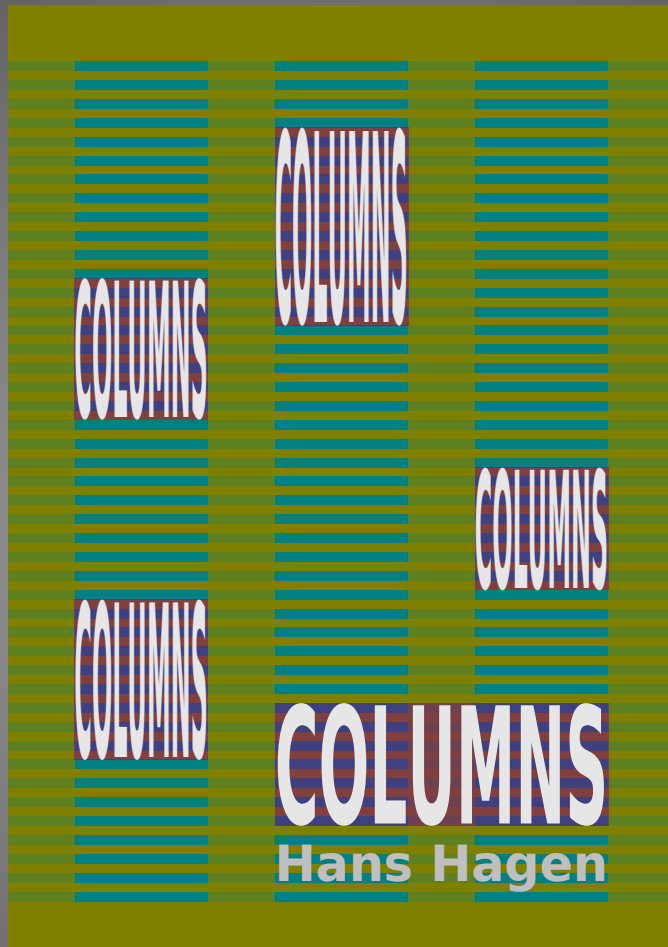
This document keeps track of the development history of both ConT<sub>E</sub>Xt (mkiv) as well as LuaT<sub>E</sub>X from the moment we considered ourselves to be halfway in the project. Like the MK document it is also one of our torture tests.

Many of the chapters of MK were first published as articles and the same is true for this document. So, the version published on the web lags behind as we don't want to compete with the user group journals.



The swiglib project provides a framework for using additional Lua libraries in Lua<sub>T</sub><sub>E</sub>X. Although Con<sub>T</sub><sub>E</sub>Xt supports this, it will not depend on external libraries for typesetting. In the meantime this subsystem has become obsolete.

The swiglib project provides a framework for using additional Lua libraries in Lua<sub>T</sub><sub>E</sub>X. Although Con<sub>T</sub><sub>E</sub>Xt supports this, it will not depend on external libraries for typesetting. In the meantime this subsystem has become obsolete.



## Columnsets (old)

Column sets can be used for quite complex but nice looking layouts. They are used for magazine like layouts and mix well with explicitly placed graphics. The MkIV version is a bit different from the MkII version but uses the same principles.



# Math

This a preliminary manual about some aspects of math typesetting. It is not a replacement for the Knuthian references.

# Spacing

This a work-in-progress manual about aspects of spacing in ConT<sub>E</sub>Xt MkIV.

Spacing in ConT<sub>E</sub>Xt

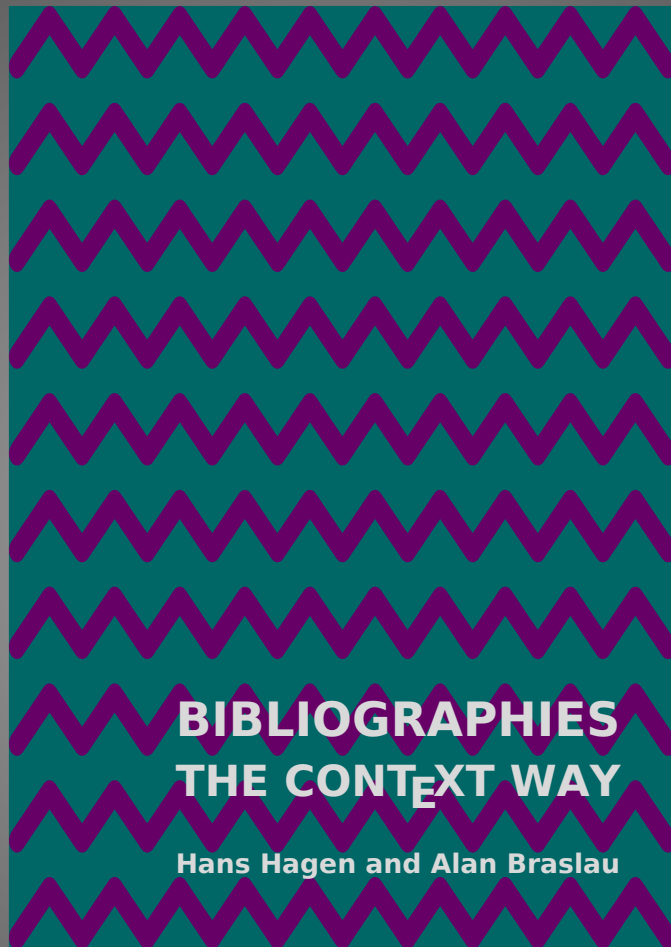


workflow  
support in context

## Workflows

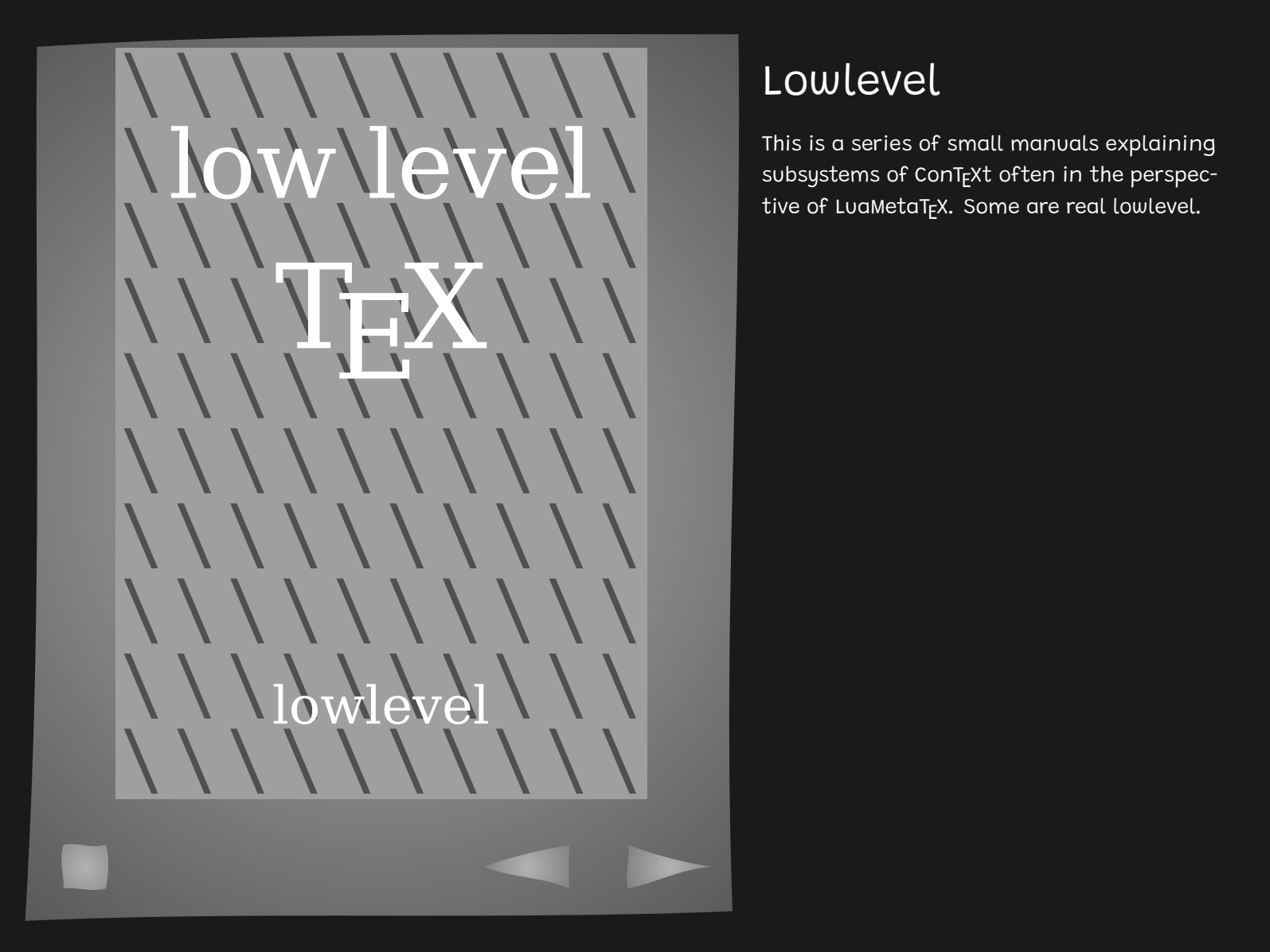
The ConT<sub>E</sub>Xt ecosystem is of course centered around typesetting but in addition comes with all kind of tools and subsystems for managing the process. Here we collect some tips.





## Publications

In ConT<sub>E</sub>Xt we support the bibT<sub>E</sub>X format for handling references. The subsystem for dealing is flexible enough to deal with many situations and is extensible as well. It does not depend on external tools and is driven by Lua on the one hand and ConT<sub>E</sub>Xt setups on the other.



low level

TEX

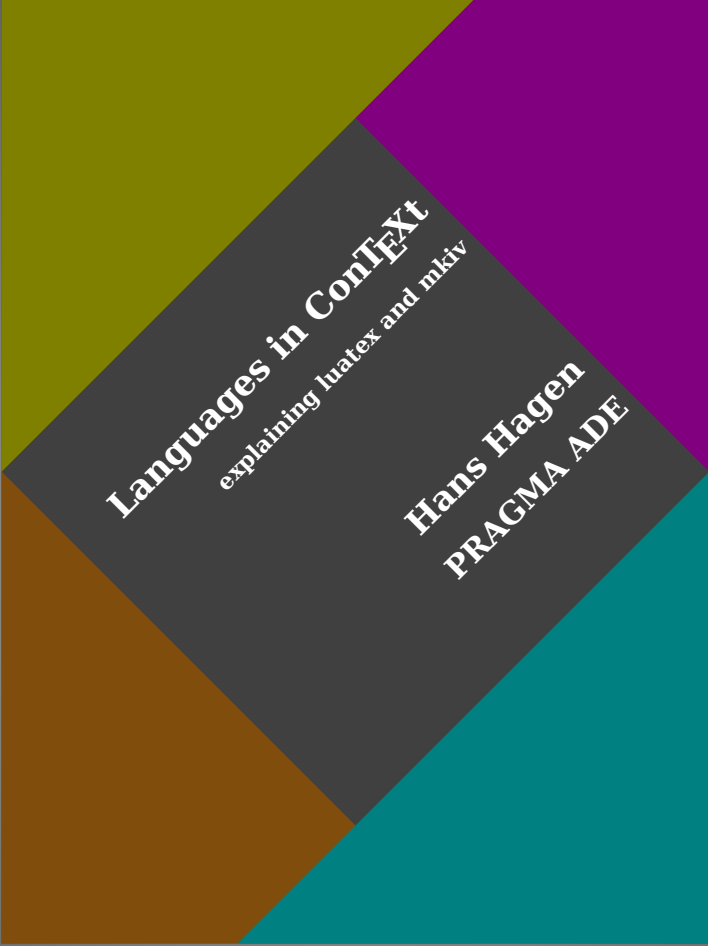
lowlevel

## Lowlevel

This is a series of small manuals explaining subsystems of ConTEXt often in the perspective of LuaMetaTEX. Some are real lowlevel.

# Languages

The ability to deal with many languages is an important property of T<sub>E</sub>X systems. Here we cover aspects like hyphenation and language dependent labels.

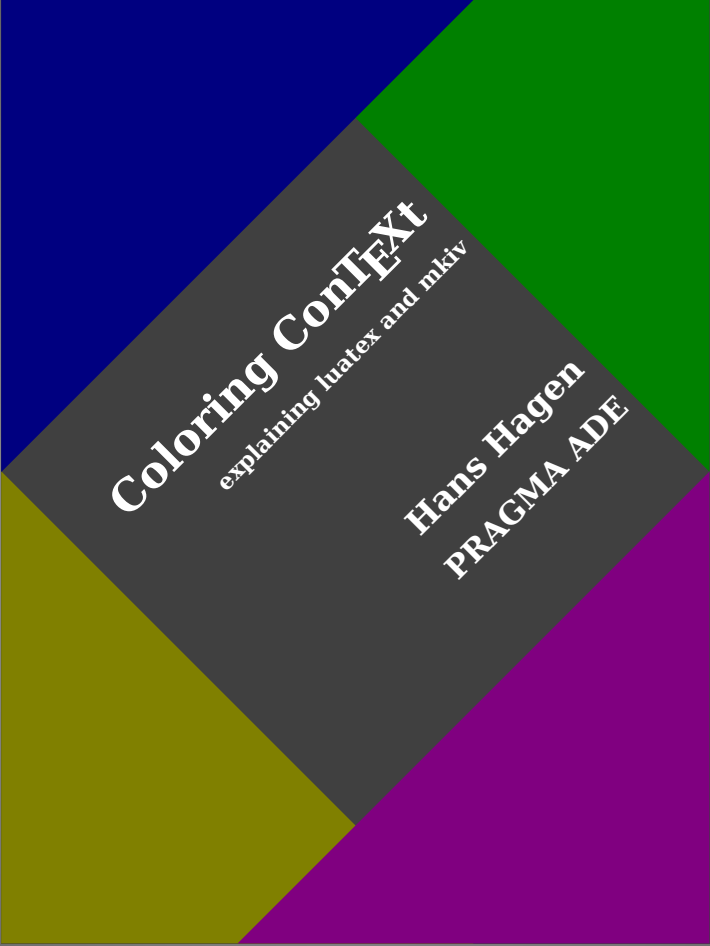


The book cover features a central dark gray diamond shape. The top-left corner of the cover is olive green, the top-right is magenta, the bottom-left is brown, and the bottom-right is teal. The title and author's name are printed in white on the gray diamond.

**Languages in Context**  
explaining luatex and mkiv

**Hans Hagen**  
**PRAGMA ADE**





# Coloring ConTeXt

explaining luatex and mkiv

Hans Hagen  
PRAGMA ADE

## Colors

Color support is like fonts and languages a core mechanism. This manual is part of the more technical description of features like that.

hans hagen

# about

luatex and context

## About LuaT<sub>E</sub>X and ConT<sub>E</sub>Xt

This is the third document in the series about the development of LuaT<sub>E</sub>X and MkIV. This one goes under the name ‘about’ as one might wonder what all this development is about. After all we’ve now reached a state where we can think about future applications instead of improving older features as that process is ongoing.

As we’re a bit beyond experimenting now, the focus will be on practical usage and of course we target on applications that the Lua and T<sub>E</sub>X combination makes possible, either new or in a renewed form.

HANS HAGEN, PRAGMA ADP, JULY 24, 2007

# DEALING WITH XML IN CONTEXT MkIV

## Dealing with XML

This manual explains how to define styles for tree based processing of xml files. This variant showed up in MkIV. The manual also contains examples of filtering content.

# ConTeXt Lua Documents

Hans Hagen

## ConTeXt Lua Documents

This manual describes how to generate documents (structure as well as content) using Lua exclusively. Of course you can also embed such code in your normal  $\TeX$  documents but using Lua has some advantages when you deal with for instance database output.



# luatools mtxrun context

## Luatools, Mtxrun & Context

Here we discuss the main tools on the Con-  
T<sub>E</sub>Xt suite of programs. We focus on the lua-  
tools tree handler, the mtxrun script man-  
ager and the process management tool Con-  
T<sub>E</sub>Xt.



**HANS HAGEN**  
**CONTEXT MKIV**

# EXTREME TABLES

## Extreme Tables

This is a short introduction to yet another table mechanism built in ConT<sub>E</sub>Xt. It is a variant of the so called natural tables but it has a different configuration. These tables are faster to process and can span lots of pages.

**HANS HAGEN**  
**CONTEXT MKIV**

# S T I N G

## Units

As part of physics support the ConT<sub>E</sub>Xt core provides a mechanism for typesetting units. This manual describes the basics as well as explains how additional units can be added and extra variants of the command can be configured. The manual also introduces the related digits mechanism. The built in support for units should not be confused with the older (incompatible but conceptually similar) units module.

|     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| A01 | B01 | C01 | D01 | E01 | F01 | G01 | H01 | I01 | J01 | K01 | L01 |
| A02 | B02 | C02 | D02 | E02 | F02 | G02 | H02 | I02 | J02 | K02 | L02 |
| A03 | B03 | C03 | D03 | E03 | F03 | G03 | H03 | I03 | J03 | K03 | L03 |
| A04 | B04 | C04 | D04 | E04 | F04 | G04 | H04 | I04 | J04 | K04 | L04 |
| A05 | B05 | C05 | D05 | E05 | F05 | G05 | H05 | I05 | J05 | K05 | L05 |
| A06 | B06 | C06 | D06 | E06 | F06 | G06 | H06 | I06 | J06 | K06 | L06 |
| A07 | B07 | C07 | D07 | E07 | F07 | G07 | H07 | I07 | J07 | K07 | L07 |
| A08 | B08 | C08 | D08 | E08 | F08 | G08 | H08 | I08 | J08 | K08 | L08 |
| A09 | B09 | C09 | D09 | E09 | F09 | G09 | H09 | I09 | J09 | K09 | L09 |
| A10 | B10 | C10 | D10 | E10 | F10 | G10 | H10 | I10 | J10 | K10 | L10 |
| A11 | B11 | C11 | D11 | E11 | F11 | G11 | H11 | I11 | J11 | K11 | L11 |
| A12 | B12 | C12 | D12 | E12 | F12 | G12 | H12 | I12 | J12 | K12 | L12 |
| A13 | B13 | C13 | D13 | E13 | F13 | G13 | H13 | I13 | J13 | K13 | L13 |
| A14 | B14 | C14 | D14 | E14 | F14 | G14 | H14 | I14 | J14 | K14 | L14 |
| A15 | B15 | C15 | D15 | E15 | F15 | G15 | H15 | I15 | J15 | K15 | L15 |
| A16 | B16 | C16 | D16 | E16 | F16 | G16 | H16 | I16 | J16 | K16 | L16 |

ConTeXt MkIV

Hans Hagen

May 12, 2016

Simple Spreadsheets

# Simple Spreadsheets

This module provides an easy way to add calculations to a document in a tabular form. It is not a replacement for a decent spreadsheet program but fits well into regular document processing as done by ConTeXt.



## LMX templates

Templates as described here can be used to construct ConT<sub>E</sub>Xt documents using a more programmatic approach. The method discussed will stay but might get extended. This mechanism also introduces two new dialects: MkIX and MkXI.



# Lua

the context libraries

## Lua libraries

This manual describes how to use generic Lua modules outside ConT<sub>E</sub>Xt. The helper functions themselves are discussed in the cld manuals.



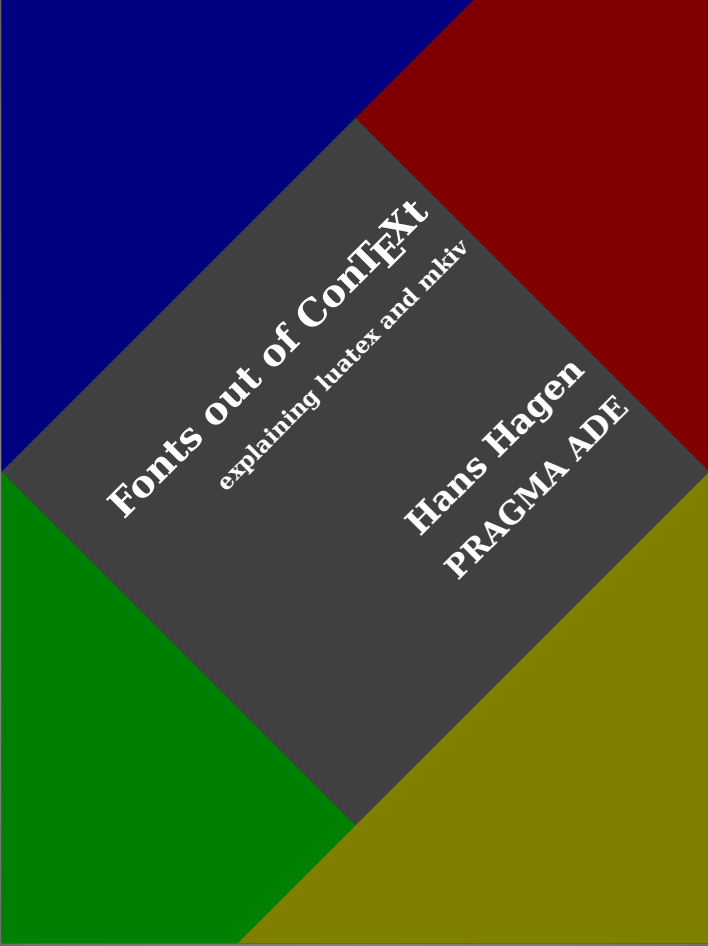
# SQL CONTEXT

## SQL in context

The ConT<sub>E</sub>Xt infrastructure can be quite handy to process sql output. This manual describes how integrate mysql support into your styles. The libraries can also be used independent from ConT<sub>E</sub>Xt but fit into the package.

# Fonts out of context

In  $\text{T}_{\text{E}}\text{X}$  and therefore in  $\text{ConT}_{\text{E}}\text{Xt}$  fonts play an important role. This document describes some of the characteristics of the  $\text{MkIV}$  font system. It is not a manual about using fonts, although some details can be found only here.



**Fonts out of Context**  
explaining luatex and mkiv

**Hans Hagen**  
**PRAGMA ADE**



# Math

This manual describes a few aspects of typesetting mathematics in ConT<sub>E</sub>Xt and will evolve over time.

<div />  
<div >  
</div>

**exporting  
xml and epub  
from context**

## Epub

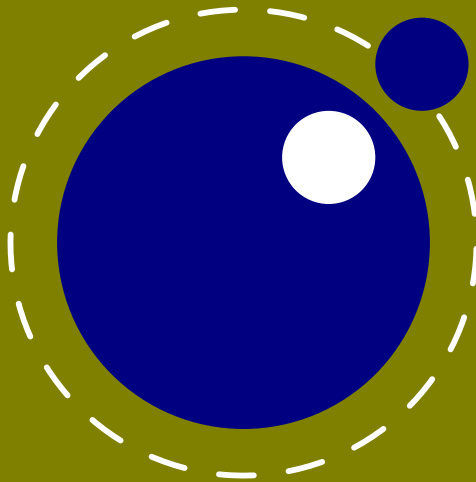
The export option in ConT<sub>E</sub>xt can produce a basic set of xml and epub files that can be either used directly (using a css) or enhanced for usage otherwise. This manual gives an overview of the process.



# **LuaT<sub>E</sub>X**

## **Reference**

### **Manual**

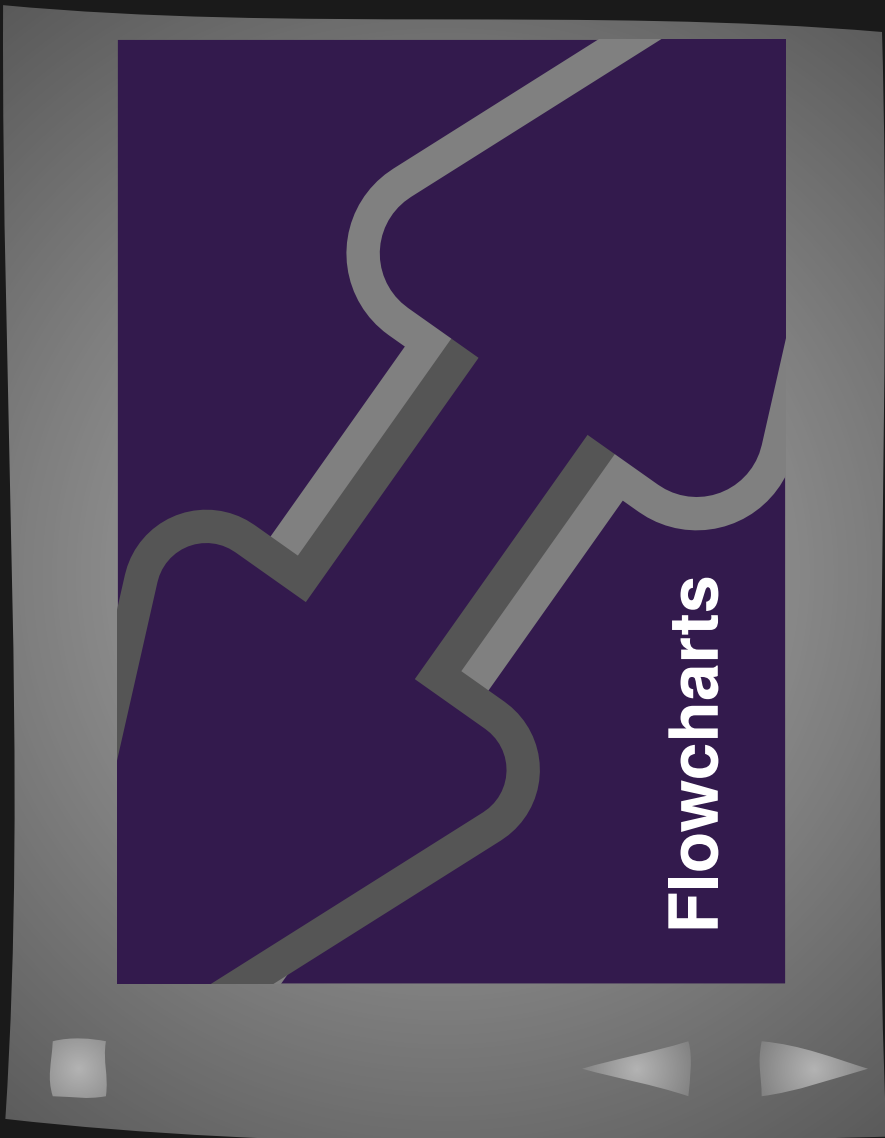


stable  
July 2021  
Version 1.10

## **LuaTeX**

The MkIV version of ConT<sub>E</sub>Xt uses the LuaT<sub>E</sub>X engine. This engine is an ongoing development and happens in the scope of ConT<sub>E</sub>Xt development. This manual describes the current version and is offered here for convenience.





# Flowcharts

## Flowcharts

The flowchart module is an old one that has been around for a while. It got updated to MkIV and wil stay around.





ConTEXt  
lmtx

FOLLOWING UP

## Followingup

This is the fifth document in the series about the development of LuaTEX and MkIV, but here we focus on LuaMetaTEX instead. It is a relative small set of progress reports that describes the steps to get there. This engine is a follow up on LuaTEX, the engine that is used for the ConTEXt version tagged as LMTX. While the functionality of its ancestor is more or less set in stone, so that it can be used in other macro packages, this follow up permits further experiments in ConTEXt.

# RULES

HANS HAGEN  
A CONTEXT MKIV MANUAL

## Rules

In this manual we cover some aspects of drawing (ornamental) rules in ConT<sub>E</sub>Xt using native rule operators as well as MetaPost.

l2r

r2l

a few tips

hans hagen

## Bidi

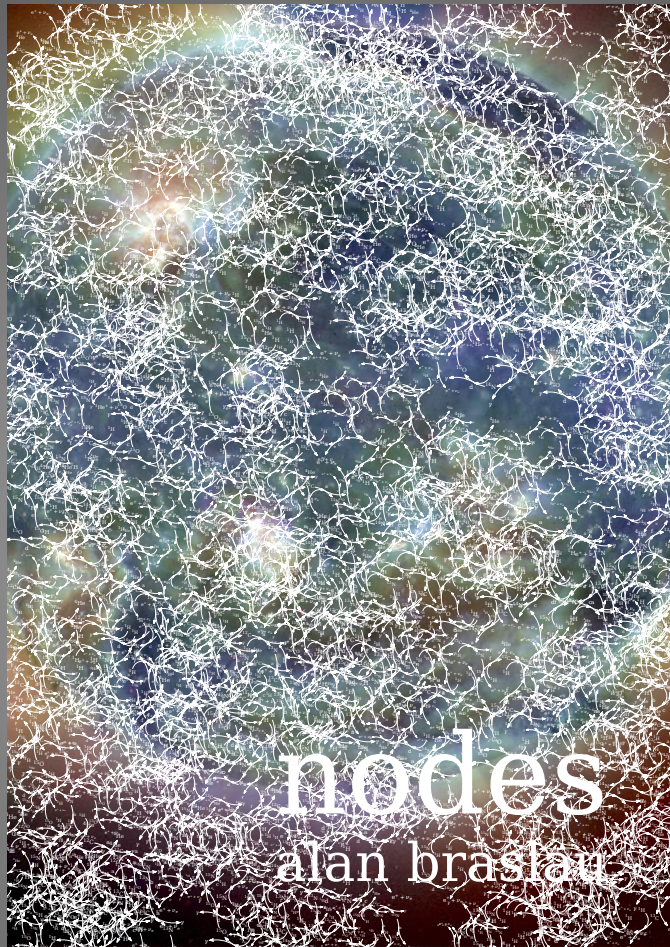
Right to left typesetting involves directives, fonts, heuristics, and a sometimes dedicated layout. In ConT<sub>E</sub>Xt some mechanism are direction aware. Here we discuss some details.

# Context<sup>t</sup> Musings

hanS hagen

## Musings

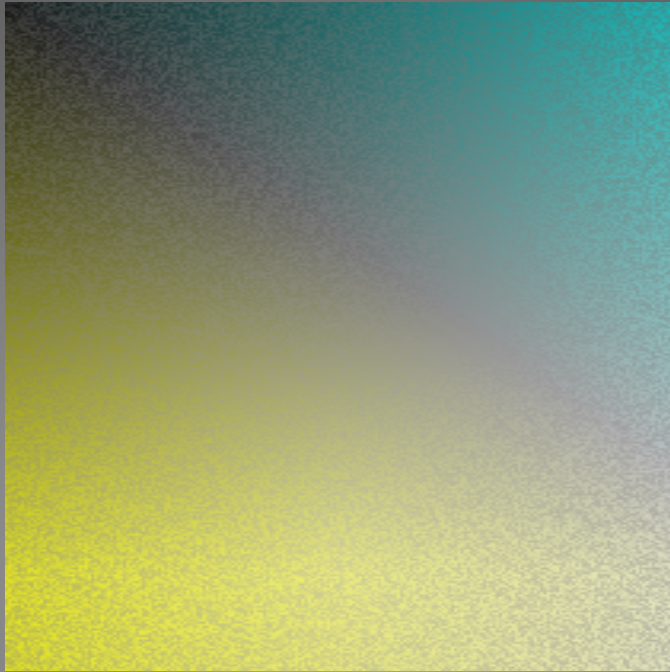
In this manual we collect articles that don't fit into another manual or collection. Some relate to talks, other to experiences or observations. They are often opinions.



# Nodes

This manual is about a rather neat set of macros to produce node related drawings in MetaPost and ConT<sub>E</sub>Xt like charts and trees. It also presents some tricks that can be applied elsewhere.





# Graphics

Hans Hagen

## Graphics

This manual explains how to insert images into a document.



The logo for PDFmerge, featuring the letters 'P', 'D', and 'F' repeated in a grid-like fashion. The letters are white and set against a dark olive green background. The 'P' and 'D' are larger and more prominent, while the 'F' is smaller and positioned to the right.

merging, embedding, fixing  
and messing a bit around  
in context lmtx

## PDFmerge

This manual explains how to efficiently include pages from pdf files into documents. It also goes into detail about some cleanup and manipulation of the content.



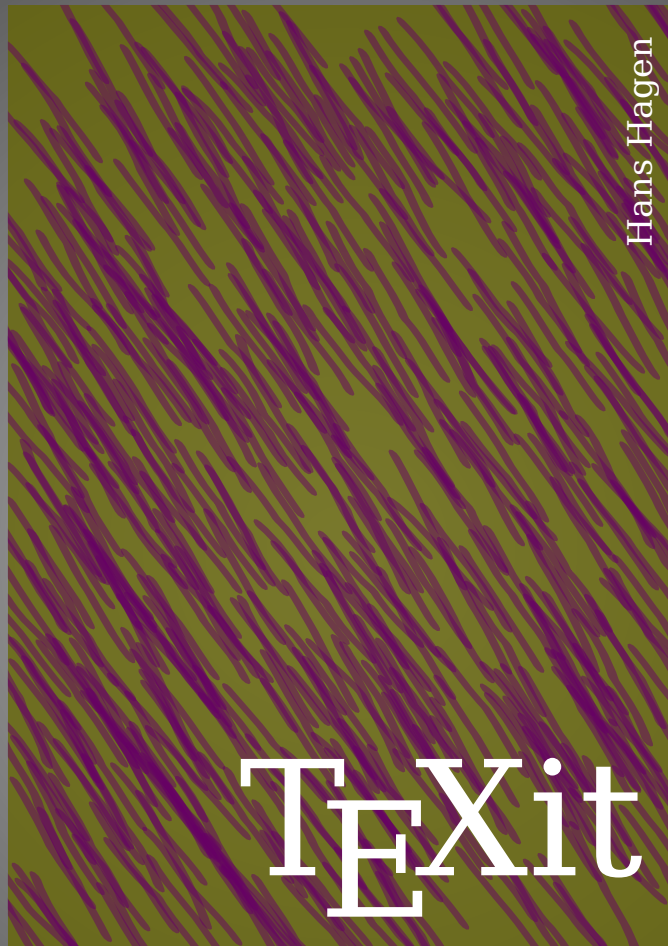
# Still going on

Hans Hagen

## Still

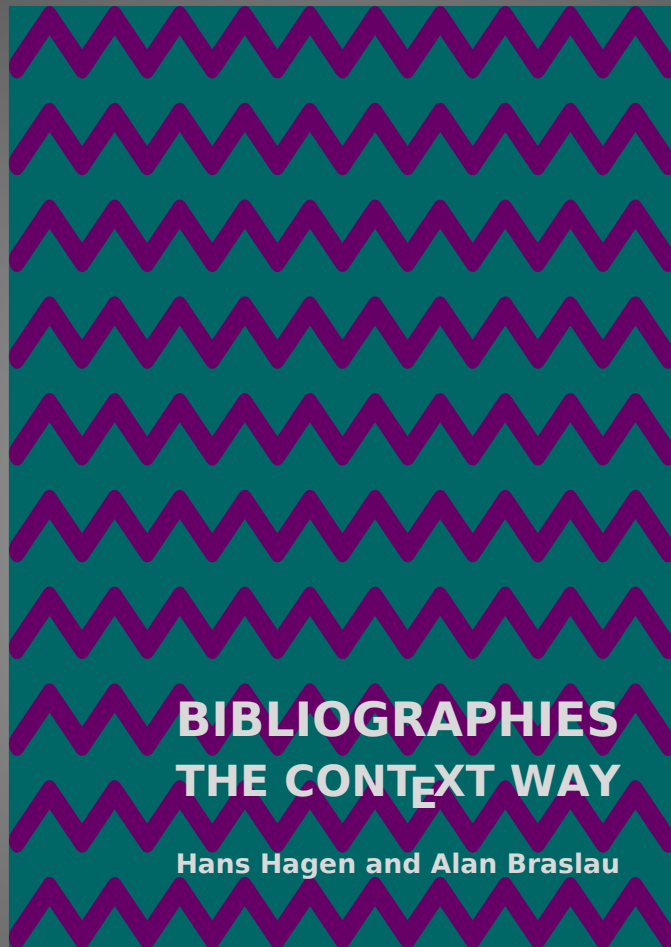
This is the fourth document describing the history of Lua $\TeX$ . Most of the development is done, but we keep on playing with the possibilities it offers for Con $\TeX$ t. We finally arrive at version 1.0 too.





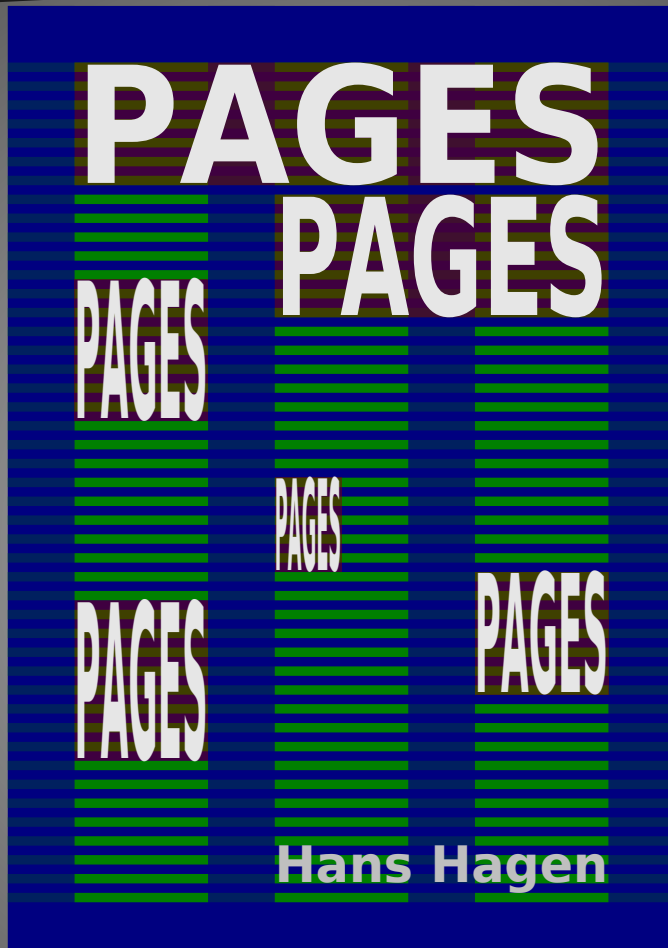
## TeXit

Sometimes questions on the mailing list pop up that demand a bit more tex-nical explanation. This manual will collect explanations and insights that don't fit into regular manuals.



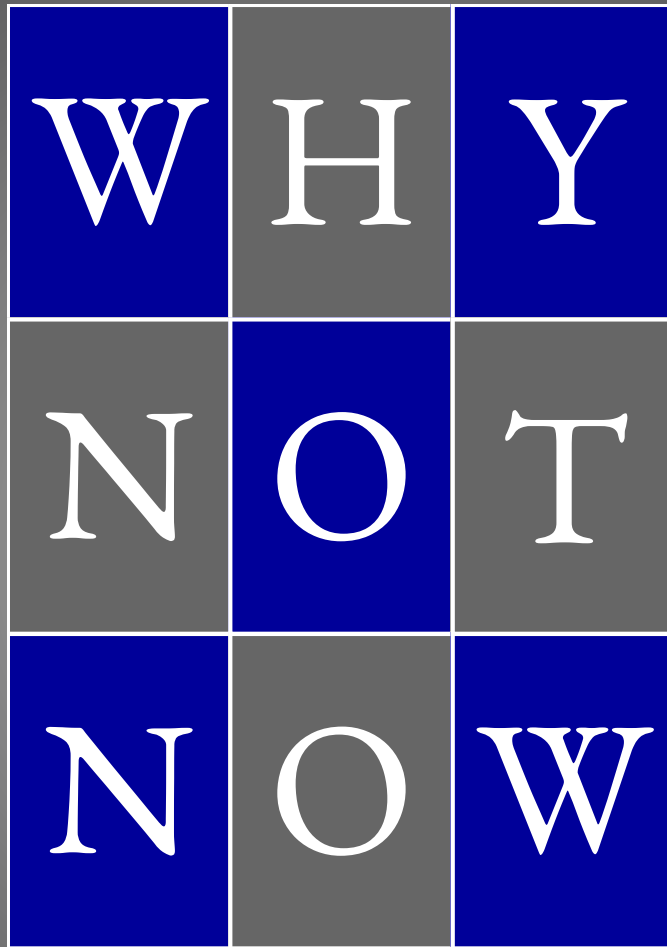
## Publications

Like any macro package ConT<sub>E</sub>Xt has to support bibliographies. This manual describes in great detail how to copye with this, and especially bibT<sub>E</sub>X databases and finetuning the rendering. We got rid of dependencies of external programs and all happens in Lua. This also opens up access to the data to users for various purposes.



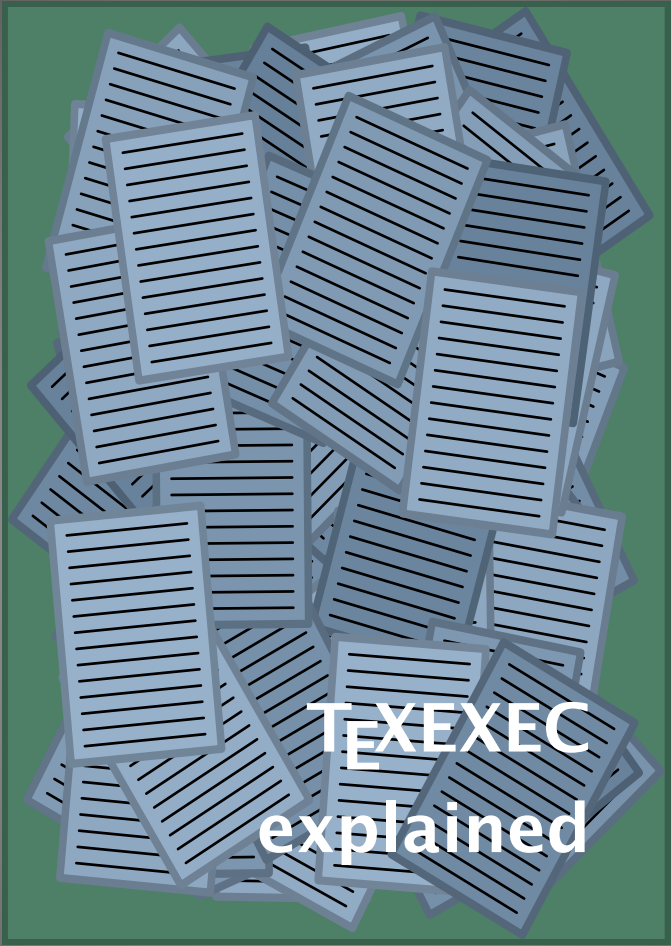
## Page columns

There are several column mechanisms and this is one of them. It boils down to treating each column as a page which in turn means that we can do for instance side floats. This manual might also give you an idea about its usability.



## Not now

This is more an excuse manual: why are some features not supported or limited.



# $\text{T}_{\text{E}}\text{XEXEC}$ explained

## $\text{T}_{\text{E}}\text{Xexec}$

Traditional  $\text{T}_{\text{E}}\text{X}$  is hard to control on the commandline. This is why Con $\text{T}_{\text{E}}\text{Xt}$  comes with  $\text{T}_{\text{E}}\text{Xexec}$ , a Perl script that makes document processing more convenient. This script also helps you to postprocess pdf files, typeset Con $\text{T}_{\text{E}}\text{Xt}$  documentation, arrange pages, and manage files.

# Typographic Programming

Designing styles is a mixture of making the right decisions in setting up the layout, finding the right values for the parameters that determine the typographic quality of the paragraph and page, and writing programs that take care of constructing the special elements that make up the page. This document tries to provide some insight in these matters and will be completed when we have time or reason.

Typographic  
Programming

Hans Hagen

# Fonts in ConT<sub>E</sub>Xt

Hans Hagen  
October 2001

## Fonts

Although installation of T<sub>E</sub>X and friends has become relatively easy, fonts always will be a special case. This is a result from the flexibility of T<sub>E</sub>X, as well as the fact that T<sub>E</sub>X can typeset virtually any language.

The font manual covers the installation of fonts in ConT<sub>E</sub>Xt and describes in detail how to define typescripts, how to achieve special effects, like hanging punctuation, and how to set up math fonts.

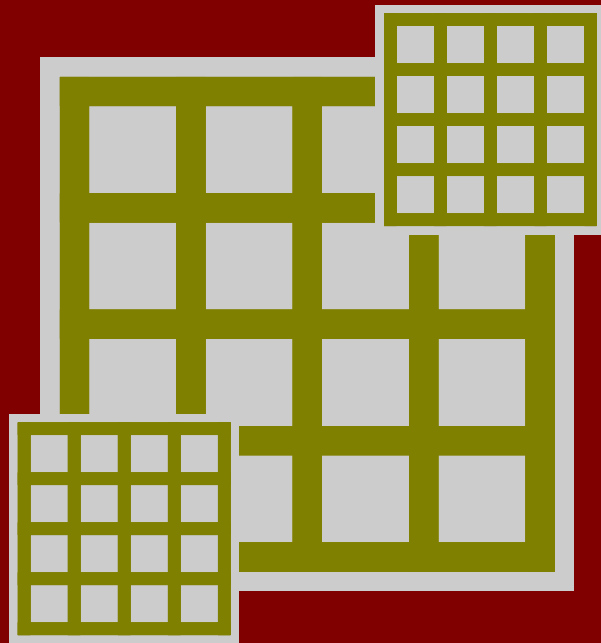
showfont

showunic

```
name:
documents/general/manuals/mkiv

file:
documents/general/manuals/mkiv

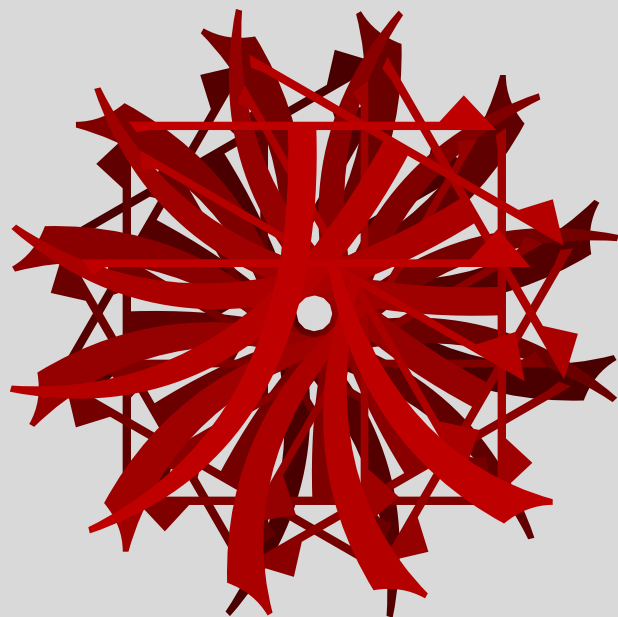
state: unknown
```



# **T<sub>E</sub>XFONT** **explained**

## Fonts

Installing fonts is one of the nasty parts of using T<sub>E</sub>X. This is why ConT<sub>E</sub>Xt comes with a Perl script called T<sub>E</sub>Xfont. You can use T<sub>E</sub>Xfont to generate font metric files in specified encodings, manipulate fonts, creating instances of multiple master fonts, build map files, etc. The script runs on top of afm2t<sub>f</sub>m and the mminstance tools.



## Chinese in ConT<sub>E</sub>Xt

## Chinese

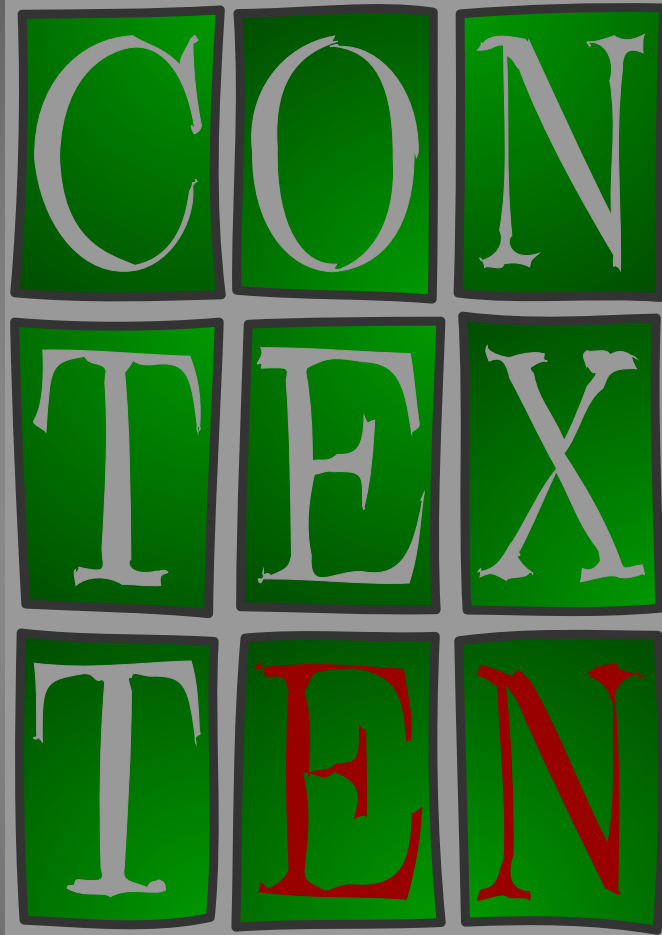
In many aspects, typesetting Chinese differs from typesetting Latin languages. Most noticeably are the pictographic characters, vertical typesetting, multiple numbering systems, and a different way of handling labels.

This manual covers the specific font setups, encoding issues, and mixed Latin and Chinese typesetting.



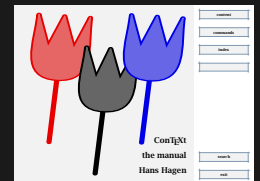
# What is ConTeXt

Occasionally I run into a description of ConTeXt that contains observations that are somewhat off. This document provides some insight in why this macro package looks the way it looks. What started out as a T<sub>E</sub>X only system evolved via adding MetaPost to the current hybrid system that also uses Lua.



# The Manual

This is the big reference manual, the one that is supposed to cover the whole of Con-<sub>T</sub>EXT. This manual is written for MkII but a lot of it still applies to MkIV. Especially fonts, encodings and languages are different in MkIV. For most commands the user interface hasn't changed, so don't be fooled by the fact that this manual is old.



# PPCHTEX

a macropackage for typesetting  
chemical structure formulas

J. Hagen & A.F. Otten  
Pragma ADE, Hasselt NL  
November 2001

## Chemistry

ppchTeX is a relatively independent macro package that can be used to typeset chemical formulas. These manuals show how it's done. There were also some faq's and a suite with many examples but we removed these.



# MATHML

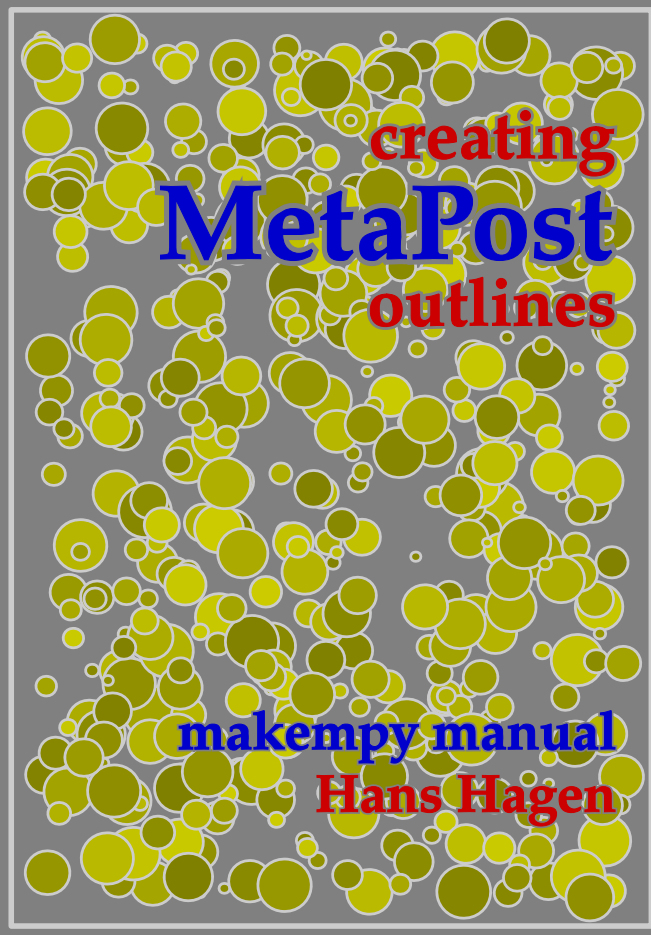
## ConT<sub>E</sub>Xt XML

Pragma ADE / Hasselt NL

## MathML support

This (short) manual explains how to invoke MathML support in ConT<sub>E</sub>Xt. It can be seen as an addendum to the MathML manual.





## MetaPost outlines

MakeMPY is a Perl script and some macros that make it possible to create outlines from text typeset by  $\text{T}_\text{E}\text{X}$ , that can be imported into MetaPost graphics. This toolkit uses pdf $\text{T}_\text{E}\text{X}$ , pdftops, pstoeit and Ghostscript, and works with any  $\text{T}_\text{E}\text{X}$ .



# XML in ConT<sub>E</sub>Xt

introduction  
general markup  
processing files  
defining interfaces  
basic workflows  
some examples  
command reference

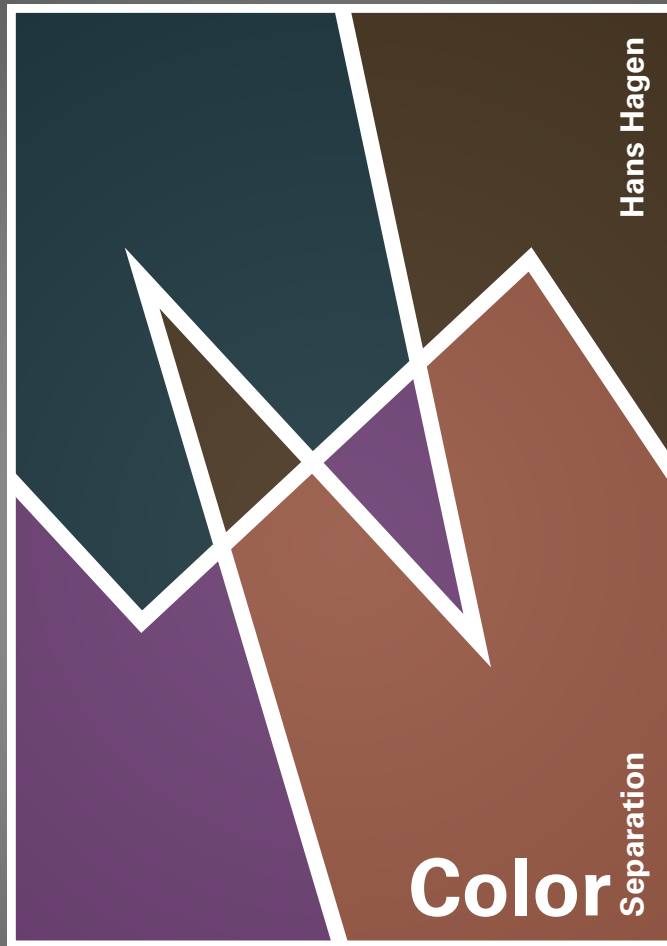
PRAGMA ADE | May 9, 2002

exit begin reference

< > < > < > < > < > < >

## XML

Since T<sub>E</sub>X can handle ascii input rather well, it will be no surprise that ConT<sub>E</sub>Xt can handle xml. In this document we describe the interface to xml. We also provide some examples, tips and tricks. This document is still under construction.

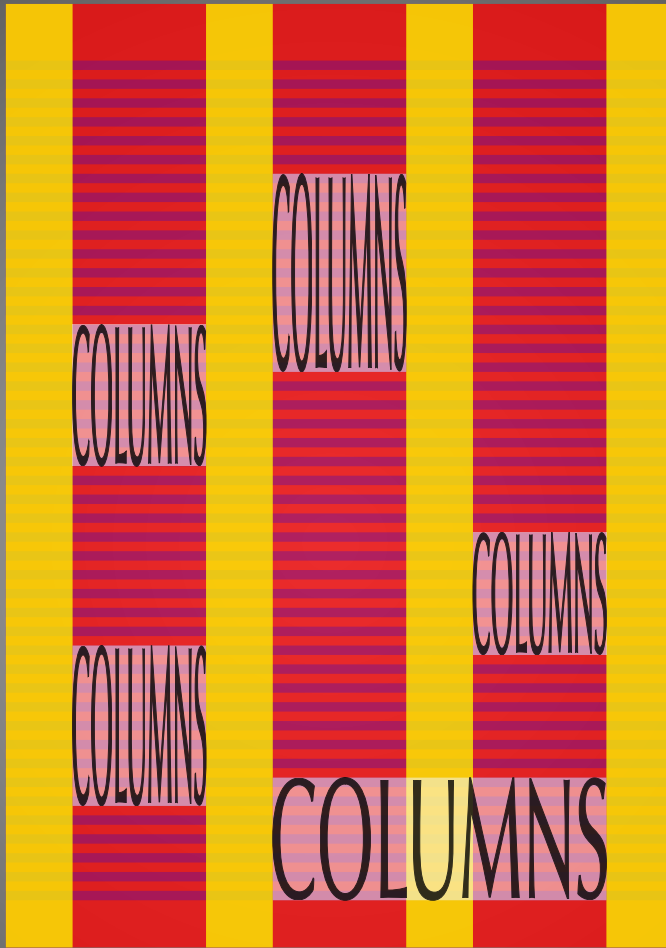


# Color separation

This is a manual for those who are forced to deliver their typeset results color separated. The manual describes how to create an instance of a document in a specific color space and channel. Text as well as graphics are covered.

# Extreme Columns

Column sets can be used for quite complex but nice looking layouts. (Behind the screens) this mechanism goes to the extremes of what we can do with  $\text{\TeX}$ 's output routines. With columnsets we try to bridge between sequential makeup and semi automated desk top publishing.





# CHARTS uncovered

## Charts

The flow chart module is an example of combining the power of  $\text{T}_{\text{E}}\text{X}$  and MetaPost. You can use this module to define charts in a descriptive way such that parts can be used, and or charts can be combined. The advantage of using this integrated approach (opposite to dedicated programs) is that you have the whole Con $\text{T}_{\text{E}}\text{X}$ t machinery available, like hyperlinks and embedded graphics. Also, by using this module, you have a proper match of fonts between graphics and text.



## Weaving PS into PDF

This manual describes the `pstopdf` tool that comes with `ConTEXt`. You can use this tool to convert PostScript images into pdf. The program is actually a wrapper around Ghostscript, but applies some additional trickery and filtering. It also supports watched folders and is suited for interfacing to the `eXaMpLe` framework.



# TEXMFSTART

Hans Hagen – 2003/2006

## texmfstart & ...tools

This very short manual demonstrates how you can use texmfstart to launch scripts and documents located in your  $\TeX$  tree. The script uses kpsewhich as well as its own (more aggressive) methods for locating the file.

The  $\TeX$ tools manual describes a program that actually is a (growing) collection of small utilities that operate on  $\TeX$  related files and trees.

The xmltools manual describes a similar program, this time a collection of utilities that operate on cq. produce xml files and trees.

Finally, the pdftools manual deals with the associated program, that operates on pdf files. This tool is not yet public.

Hans Hagen

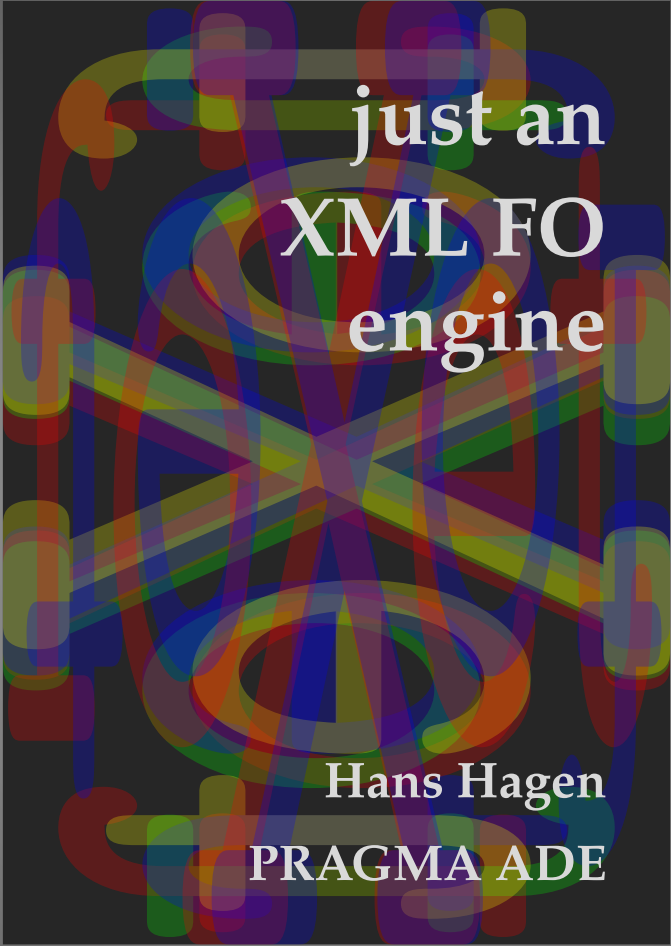
PATTERNS

HYPHENATION

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| AA | AB | AC | AD | AE | AF | AG | AH | AI | AJ | AK | AL | AM | AN | AO | AP | AQ | AR | AS | AT | AU | AV | AW | AX | AY | AZ |
| BA | BB | BC | BD | BE | BF | BG | BH | BI | BJ | BK | BL | BM | BN | BO | BP | BQ | BR | BS | BT | BU | BV | BW | BX | BY | BZ |
| CA | CC | CD | CE | CF | CG | CH | CI | CJ | CK | CL | CM | CN | CO | CP | CQ | CR | CS | CT | CU | CV | CW | CX | CY | CZ |    |
| DA | DC | DD | DE | DF | DG | DH | DI | DJ | DK | DL | DM | DN | DO | DP | DQ | DR | DS | DT | DU | DV | DW | DX | DY | DZ |    |
| EA | EC | ED | EE | EF | EG | EH | EI | EJ | EK | EL | EM | EN | EO | EP | EQ | ER | ES | ET | EU | EV | EW | EX | EY | EZ |    |
| FA | FC | FD | FE | FF | FG | FH | FI | FJ | FK | FL | FM | FN | FO | FP | FQ | FR | FS | FT | FU | FV | FW | FX | FY | FZ |    |
| GA | GC | GD | GE | GF | GG | GH | GI | GJ | GK | GL | GM | GN | GO | GP | GQ | GR | GS | GT | GU | GV | GW | GX | GY | GZ |    |
| HA | HC | HD | HE | HF | HG | HH | HI | HJ | HK | HL | HM | HN | HO | HP | HQ | HR | HS | HT | HU | HV | HW | HX | HY | HZ |    |
| IA | IC | ID | IE | IF | IG | IH | II | IJ | IK | IL | IM | IN | IO | IP | IQ | IR | IS | IT | IU | IV | IW | IX | IY | IZ |    |
| JA | JC | JD | JE | JF | JG | JH | JI | JJ | JK | JL | JM | JN | JO | JP | JQ | JR | JS | JT | JU | JV | JW | JX | JY | JZ |    |
| KA | KC | KD | KE | KF | KG | KH | KI | KJ | KK | KL | KM | KN | KO | KP | KQ | KR | KS | KT | KU | KV | KW | KX | KY | KZ |    |
| LA | LC | LD | LE | LF | LG | LH | LI | LJ | LK | LL | LM | LN | LO | LP | LQ | LR | LS | LT | LU | LV | LW | LX | LY | LZ |    |
| MA | MC | MD | ME | MF | MG | MH | MI | MJ | MK | ML | MM | MN | MO | MP | MQ | MR | MS | MT | MU | MV | MW | MX | MY | MZ |    |
| NA | NC | ND | NE | NF | NG | NH | NI | NJ | NK | NL | NO | NP | NQ | NR | NS | NT | NU | NV | NW | NX | NY | NZ |    |    |    |
| OA | OC | OD | OE | OF | OG | OH | OI | OJ | OK | OL | OM | ON | OO | OP | OQ | OR | OS | OT | OU | OV | OW | OX | OY | OZ |    |
| PA | PC | PD | PE | PF | PG | PH | PI | PJ | PK | PL | PO | PP | PQ | PR | PS | PT | PU | PV | PW | PX | PY | PZ |    |    |    |
| QA | QC | QD | QE | QF | QG | QH | QI | QJ | QK | QL | QM | QN | QO | QP | QQ | QR | QS | QT | QU | QV | QW | QX | QY | QZ |    |
| RA | RC | RD | RE | RF | RG | RH | RI | RJ | RK | RL | RM | RN | RO | RP | RQ | RR | RS | RT | RU | RV | RW | RX | RY | RZ |    |
| SA | SC | SD | SE | SF | SG | SH | SI | SJ | SK | SL | SM | SN | SO | SP | SQ | SR | SS | ST | SV | SW | SX | SY | SZ |    |    |
| TA | TC | TD | TE | TF | TG | TH | TI | TJ | TK | TL | TM | TN | TO | TP | TQ | TR | TS | TT | TU | TV | TW | TX | TY | TZ |    |
| UA | UC | UD | UE | UF | UG | UH | UI | UJ | UK | UL | UM | UN | UO | UP | UQ | UR | US | UT | UU | UV | UW | UX | UY | UZ |    |
| VA | VC | VD | VE | VF | VG | VH | VI | VJ | VK | VL | VN | VO | VP | VQ | VR | VS | VT | VU | VV | VW | VX | VY | VZ |    |    |
| WA | WC | WD | WE | WF | WG | WH | WI | WJ | WK | WL | WM | WN | WO | WP | WQ | WR | WS | WT | WU | WV | WW | WX | WY | WZ |    |
| XA | XC | XD | XE | XF | XG | XH | XI | XJ | XK | XL | XM | XN | XO | XP | XQ | XR | XS | XT | XU | XV | XW | XX | XY | XZ |    |
| YA | YC | YD | YE | YF | YG | YH | YI | YJ | YK | YL | YM | YN | YO | YP | YQ | YR | YS | YT | YU | YV | YW | YX | YY | YZ |    |
| ZA | ZC | ZD | ZE | ZF | ZG | ZH | ZI | ZJ | ZK | ZL | ZM | ZN | ZO | ZP | ZQ | ZR | ZS | ZT | ZU | ZV | ZW | ZX | ZY | ZZ |    |

# Hyphenation Patterns

Although normally users are not supposed to know the dirty details of pattern management, it may be handy to read this manual at least once, if only to know what to do when for some reason pattern loading fails on your machine. This manual also describes how to apply the `ctxtools` program to generate generic pattern files from existing encoding specific files.



# just an XML FO engine

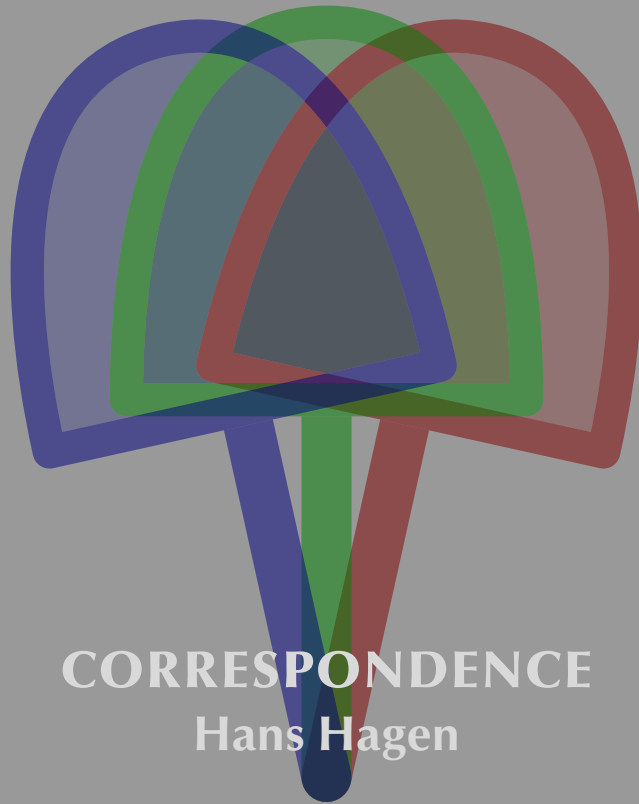
Hans Hagen  
PRAGMA ADE

## foXet

You can see foXet as just another way of processing xml formatting objects. You may use it to process documents coded in (reasonable) xsl-fo or as (textual) graphics format in ConT<sub>E</sub>Xt documents, a sort of placed xml.



foXet

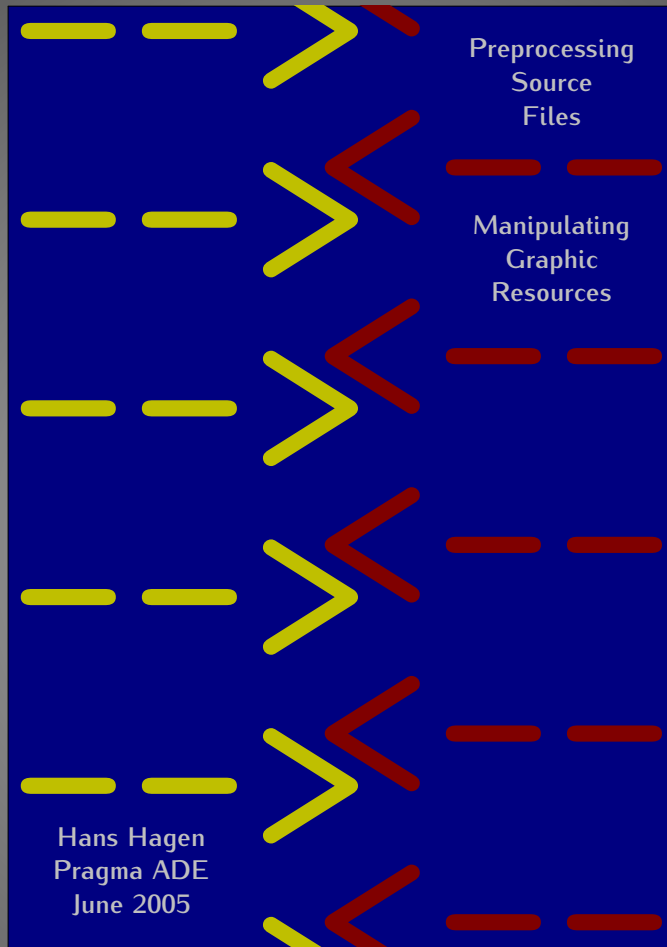


## Correspondence

One of the first application at Pragma ADEof  $\text{T}_{\text{E}}\text{X}$  was in typesetting letters. Over time the  $\text{T}_{\text{E}}\text{X}$  only based system moved to a combination of xml and  $\text{T}_{\text{E}}\text{X}$ . This manual roughly describes the components that make up such a system. A graphical user interface is provided as well.

# Preprocessing and Manipulating

This manual describes the facilities for automatic preprocessing of source files and manipulation of graphics. These features come in handy in automated typesetting workflows and are handled by  $\TeX$ exec and  $\text{rlxtools}$ . The definition files are xml based.

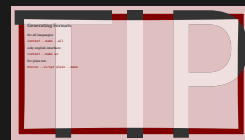


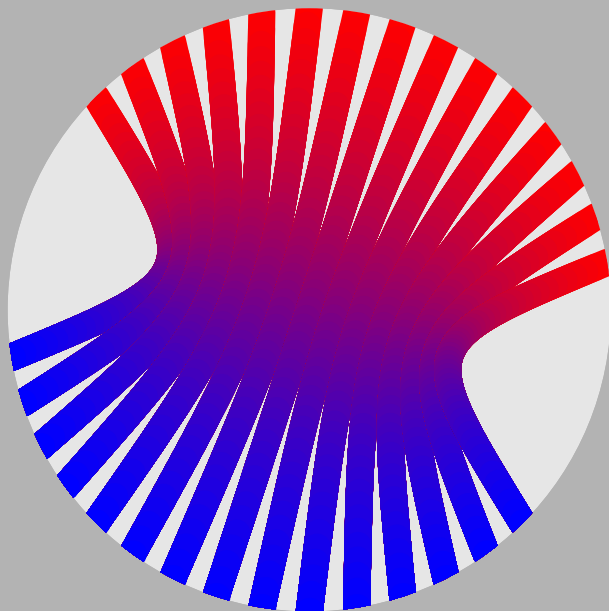


# how to install ConT<sub>E</sub>Xt

## Installation

When one uses  $\text{t\textsubscript{E}X}$ ,  $\text{f\textsubscript{P}T\textsubscript{E}X}$ ,  $\text{g\textsubscript{W}T\textsubscript{E}X}$ ,  $\text{Mik\textsubscript{T}E\textsubscript{X}}$  or  $\text{T\textsubscript{E}X Live}$ , installation of ConT<sub>E</sub>Xt is a breeze. Nevertheless, in this manual, we provide some information on installing ConT<sub>E</sub>Xt.

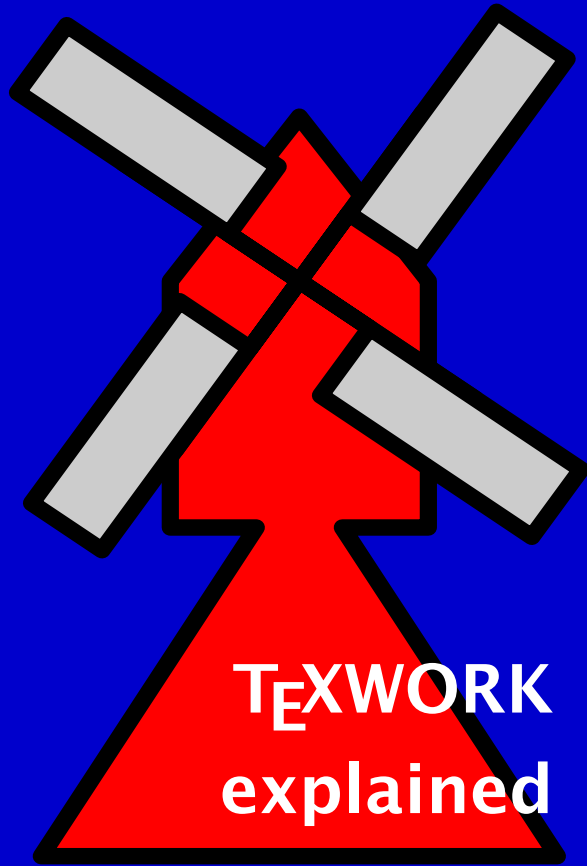




# TEXUTIL explained

## TEXutil

The TEXutil Perl script deals with files, especially the ConTEXt second pass data file. It moves information around and sorts indexes and lists. This script is the natural companion of TEXexec.

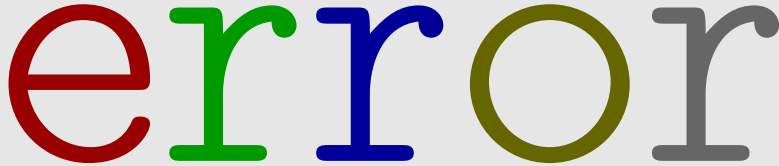


## T<sub>E</sub>Xwork

T<sub>E</sub>Xwork is our local editing environment. It is a rewrite of the Modula~2 program T<sub>E</sub>Xedit in Perl/Tk.

# Labels

The author of a graphic is not necessarily also its graphic designer. In that case it makes sense to split the design of the graphic elements from the process of adding labels. This document describes how to add text to graphics either or not using the resource (figure) library mechanism.



error



# FIGURES

ConT<sub>E</sub>Xt XML

Pragma ADE / Hasselt NL

## Figure Databases

Instead of moving hundreds of graphics around, you can package them in a database. ConT<sub>E</sub>Xt not only has means to generate such databases, but also can filter the information needed from the corresponding xml files and include graphics by label. Figure bases make it easy to swap high and low resolution graphics.



# STEPS

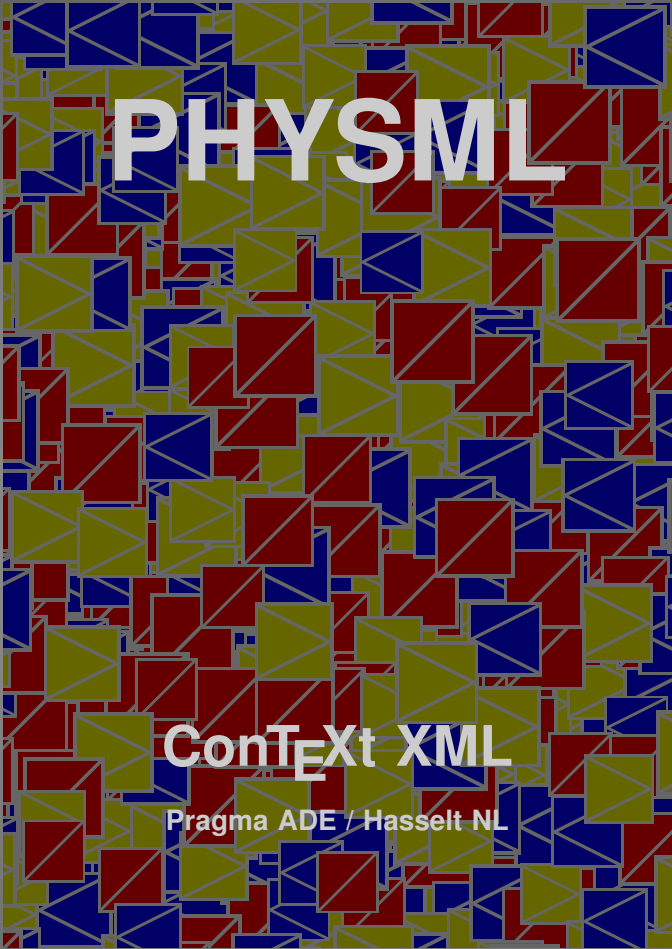
ConT<sub>E</sub>Xt XML

Pragma ADE / Hasselt NL

## Stepcharts

Stepcharts are a specific kind of tabular charts. They are a combination of MetaPost graphics and T<sub>E</sub>X code. There is a T<sub>E</sub>X as well as xml implementation.





# PHYSML

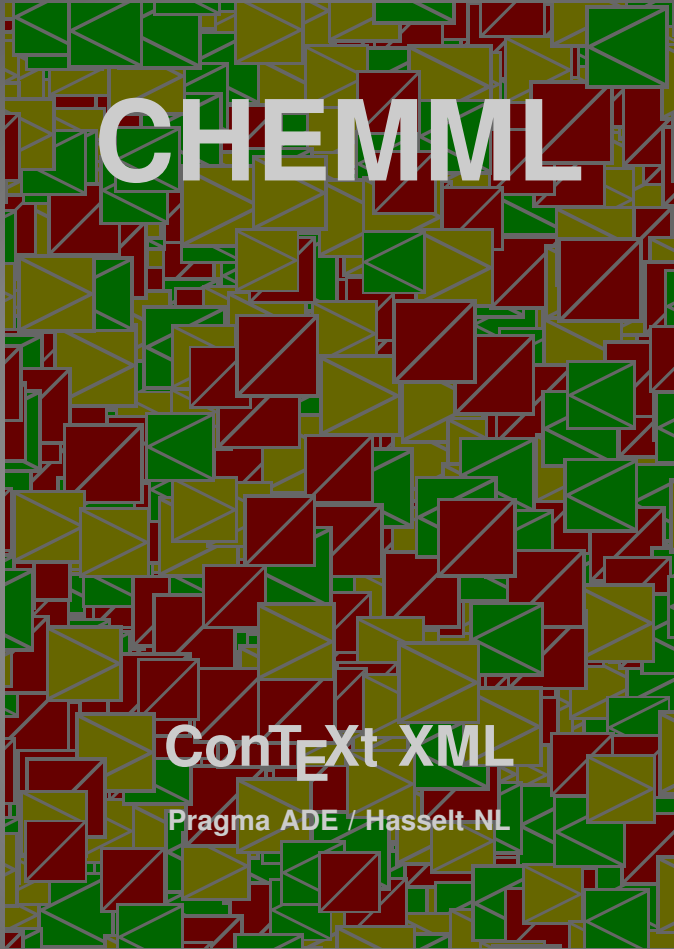
## ConT<sub>E</sub>Xt XML

Pragma ADE / Hasselt NL

## PhysML support

Support for physical units is build on top of the MathML engine. The method used is derived from the units module that comes with ConT<sub>E</sub>Xt.



The background of the slide is a dense, abstract pattern of overlapping squares and rectangles in shades of green, yellow, and red. The word 'CHEMML' is written in large, white, bold, sans-serif capital letters across the top half of the slide.

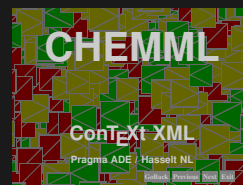
# CHEMML

ConT<sub>E</sub>Xt XML

Pragma ADE / Hasselt NL

## ChemML support

Chemical formulas have their own typographic needs. This module provides a way to code atoms, ions, molecules, and a sequence of reactions.





# WIDGETS uncovered

## Widgets

Widgets are interactive elements in (screen based) documents. This manual describes how to use the reference mechanism for advanced hyperlinking, but also discussed how to construct forms. Adding text annotations and page transitions is also discussed. This manual is replaced by the manual on interaction.

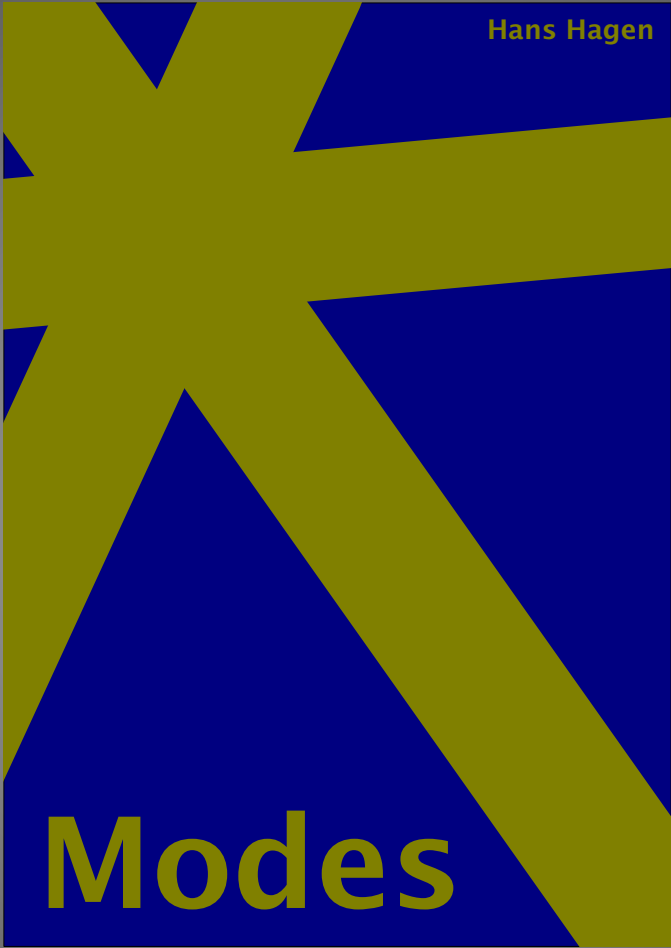
# XMT DIER

Hans Hagen – Pragma ADDE

## xmldir

This manual describes how to access information about files on your system from within ConT<sub>E</sub>Xt. The modules described here are accompanied by features in the T<sub>E</sub>Xtools script. You can use the styles to generate overviews as well as access properties of files.





Hans Hagen

# Modes

## Modes

Modes are a convenient way to create styles that serve multiple purposes. This manual describes how to enable modes and test for their state. The special system modes that `ConTeXt` sets itself are also explained.